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PROGRAM OUTLINE

Instrumentation and Control Technician (Industrial Instrument Mechanic)

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INDUSTRIAL INSTRUMENT MECHANIC

PROGRAM OUTLINE

APPROVED BY INDUSTRY

MARCH 2015

BASED ON

NOA 2013

Developed By
Industry Training Authority
Province of British Columbia

TABLE OF CONTENTS

Section 1 INTRODUCTION....4

Foreword....5
Acknowledgements....6
How to Use this Document....7

Section 2 PROGRAM OVERVIEW....9

Program Credentialing Model....10
Occupational Analysis Chart....11
Training Topics and Suggested Time Allocation....14

Section 3 PROGRAM CONTENT....20

Level 1 Industrial Instrument Mechanic....21
Level 2 Industrial Instrument Mechanic....54
Level 3 Industrial Instrument Mechanic....74
Level 4 Industrial Instrument Mechanic....102

Section 4 TRAINING PROVIDER STANDARDS....123

Instructor Requirements....124
Facility Requirements....125
Tools and Equipment....127
Reference Materials....130

Appendices....134

Appendix A Assessment Guidelines....135
Appendix B Glossary and Acronyms....140
Appendix C Sample Evaluation Sheet....145
Appendix D Previous Contributors....147

Section 1

INTRODUCTION

Industrial Instrument Mechanic

Foreword

This Program Outline is for use in the Instrumentation and Control Technician (Industrial Instrument Mechanic) apprenticeship training classes as sponsored by the Industry Training Authority and will be used as a guide for instructors in the formal classroom portions of apprenticeship training. The 2013 NOA and Alberta curriculum for this trade were examined by provincial Subject Matter Experts in the process of creating the BC 2015 Program Outline.

Practical demonstration and student participation should always be integrated with classroom sessions.

Safe working practices, though not always specified in each of the competencies and learning tasks, are an implied part of the program and should be stressed throughout the apprenticeship.

The technical training times calculated by the Industry Subject Matter Experts are based on six hours of instructional time ("student contact time") per day.

This Program Outline includes a list of recommended reference textbooks that are available to support the learning objectives and the minimum shop requirements needed to support instruction. Appendix C of this document contains a sample lab assessment tool which is intended to assist new instructors in creating lab assessment instruments.

School-based training for this trade does NOT include practical safety certification (rigging, fall protection, confined space entry, etc.). Apprentices will examine the purpose and intent of work safety documents and regulations, and know how to find this information. It is the responsibility of employers to train apprentices in on-the-job safety practices and procedures (as per BC Occupational Health and Safety Regulations and Employers' Company Safety Policies).

SAFETY ADVISORY
Be advised that references to the WorkSafeBC safety regulations contained within these materials do not/may not reflect the most recent Occupational Health and Safety Regulation (the current Standards and Regulation in BC can be obtained on the following website: http://www.worksafebc.com). Please note that it is always the responsibility of any person using these materials to inform him/herself about the Occupational Health and Safety Regulation pertaining to his/her work.

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The Industry Training Authority would like to acknowledge the dedication and hard work of all the industry and training provider representatives appointed to identify the training requirements of the Industrial Instrument Mechanic occupation.

How to Use this Document

This Program Outline has been developed for the use of individuals from several different audiences. The table below describes how each section can be used by each intended audience.

Section	Training Providers	Employers/ Sponsors	Apprentices	Challengers
Program Credentialing Model	Communicate program length and structure, and all pathways to completion	Understand the length and structure of the program	Understand the length and structure of the program, and pathway to completion	Understand challenger pathway to Certificate of Qualification
OAC	Communicate the competencies that industry has defined as representing the scope of the occupation	Understand the competencies that an apprentice is expected to demonstrate in order to achieve certification	View the competencies they will achieve as a result of program completion	Understand the competencies they must demonstrate in order to challenge the program
Training Topics and Suggested Time Allocation	Shows proportionate representation of general areas of competency (GACs) at each program level, the suggested proportion of time spent on each GAC, and percentage of time spent on theory versus practical application	Understand the scope of competencies covered in the technical training, the suggested proportion of time spent on each GAC, and the percentage of that time spent on theory versus practical application	Understand the scope of competencies covered in the technical training, the suggested proportion of time spent on each GAC, and the percentage of that time spent on theory versus practical application	Understand the relative weightings of various competencies of the occupation on which assessment is based
Program Content	Defines the objectives, learning tasks, high level content that must be covered for each competency, as well as defining observable, measureable achievement criteria for objectives with a practical component	Identifies detailed program content and performance expectations for competencies with a practical component; may be used as a checklist prior to signing a recommendation for certification (RFC) for an apprentice	Provides detailed information on program content and performance expectations for demonstrating competency	Allows individual to check program content areas against their own knowledge and performance expectations against their own skill levels

Section	Training Providers	Employers/ Sponsors	Apprentices	Challengers
Training Provider Standards	Defines the facility requirements, tools and equipment, reference materials (if any) and instructor requirements for the program	Identifies the tools and equipment an apprentice is expected to have access to; which are supplied by the training provider and which the student is expected to own	Provides information on the training facility, tools and equipment provided by the school and the student, reference materials they may be expected to acquire, and minimum qualification levels of program instructors	Identifies the tools and equipment a tradesperson is expected to be competent in using or operating; which may be used or provided in a practical assessment
Appendix – Glossary of Acronyms			Defines program specific acronyms	

Section 2

PROGRAM OVERVIEW

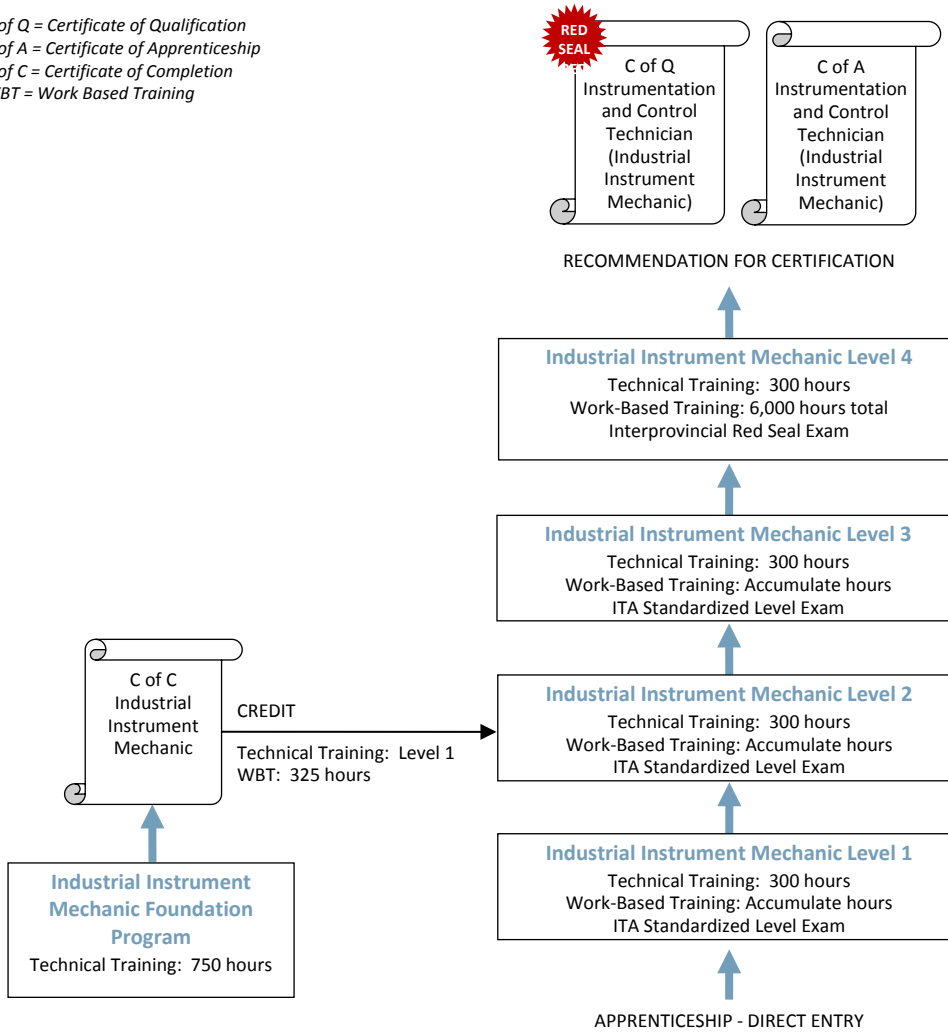
Industrial Instrument Mechanic

Program Credentialing Model

Apprenticeship pathway

This graphic provides an overview of the Instrumentation and Control Technician (Industrial Instrument Mechanic) apprenticeship pathway.

C of Q = Certificate of Qualification
C of A = Certificate of Apprenticeship
C of C = Certificate of Completion
WBT = Work Based Training



CROSS-PROGRAM CREDITS
Individuals who hold the credentials listed below are entitled to receive partial credit toward the completion requirements of this program

None

Occupational Analysis Chart

INDUSTRIAL INSTRUMENT MECHANIC

Occupation Description: "Industrial Instrument Mechanic" means a person who installs, repairs, maintains, replaces, calibrates, programs and services process monitoring and/or control instruments including indicators, recording devices, control loops and computers. These instruments may be pneumatic, hydraulic, electronic, electrical, mechanical, nuclear, optical or chemical, and include signal transmission, telemetering and digital devices in industrial operations.

OCCUPATIONAL SKILLS **A**	Plan and perform routine trade activities	A1	1					Organize work and maintain records	A2			3			Use computers and related applications	A3	1	2	3	4		Explain codes, standards and regulations	A4	1					Identify electrical hazards and apply safe work practices (includes CEC)	A5	1					Use trade related schematics and drawings	A6	1	2	3	4	
MEASURING AND INDICATING DEVICES **B**	Calibrate and service indicating and recording instruments	B1	1					Measure pressure	B2	1					Measure temperature	B3		2				Measure level	B4		2				Measure density	B5		2				Measure weight	B6		2			
	Measure flow (volumetric and mass flow)	B7		2				Measure consistency and viscosity	B8			3			Measure and service environmental monitoring devices	B9				4		Measure vibration	B10			3			Measure speed	B11			3			Measure position	B12			3		
	Measure motion	B13			3																																				
ANALYTICAL INSTRUMENTATION **C**	Measure analytical properties of process gases	C1				4		Measure analytical properties of process liquids	C2			3			Measure analytical properties of process solids	C3			3			Measure analytical properties of flue gases	C4				4																

SAFETY AND PROCESS MONITORING SYSTEMS **D**	Service and test flame safety systems D1			4		Install and service process cameras D2			4		Service ESD (emergency shutdown devices) D3			4		Service and calibrate personal safety systems D4						
PNEUMATIC AND HYDRAULIC SYSTEMS **E**	Examine air supply systems E1	1				Install tubing and fittings E2	1				Install and service pneumatic instruments E3	1		3		Install and maintain signal conditioners E4		2			Install and service control devices for hydraulic systems E5		2		
ELECTRICAL AND ELECTRONIC SYSTEMS **F**	Examine electrical theory F1	1				Apply basic principles of DC electricity F2	1				Apply basic principles of AC electricity F3	1				Apply principles of electronics F4		2			Apply Boolean logic and principles of digital electronics F5	1			
FINAL CONTROL ELEMENTS **G**	Service regulators and examine relief valves G1	1				Service, size and install control valves and actuators G2	1	2			Install and service valve positioners (includes advanced diagnostics) G3	1	2			Install and service variable speed drive (VSD) and variable frequency drive (VFD) G4			3						
COMMUNICATIONS, NETWORKING AND SIGNAL TRANSMISSION SYSTEMS **H**	Examine communication systems H1			3		Install wiring in accordance with CEC H2	1				Examine communication network structures and components H3			3		Troubleshoot signal transmission systems H4			3		Install, configure, maintain and service supervisory control and data acquisition (SCADA) systems H5			4	

CONTROL SYSTEMS I	Examine fundamental theories of process operation and equipment I1	Examine control theory I2	Examine process control techniques and strategies I3	Implement process control strategies I4	Install and service stand-alone controllers I5	Install, configure, maintain and service Distributed Control Systems (DCS) I6
		3	3	3	4	4
	Install, configure, maintain and service Programmable Logic Controllers (PLC) I7	Install, configure, maintain and service human machine interface (HMI) I8	Install and demonstrate knowledge of advanced supervisory control systems I9			
	1	3	4			

Training Topics and Suggested Time Allocation

INDUSTRIAL INSTRUMENT MECHANIC – LEVEL 1

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
Line A	**OCCUPATIONAL SKILLS**	**5%**	**80%**	**20%**	**100%**
A1	Plan and perform routine trade activities		✓	✓	
A3	Use computers and related applications		✓	✓	
A4	Explain codes, standards and regulations		✓		
A5	Identify electrical hazards and apply safe work practices (includes CEC)		✓		
A6	Use trade related schematics and drawings		✓	✓	
Line B	**MEASURING AND INDICATING DEVICES**	**15%**	**40%**	**60%**	**100%**
B1	Calibrate and service indicating and recording instruments		✓	✓	
B2	Measure pressure		✓	✓	
Line D	**SAFETY AND PROCESS MONITORING SYSTEMS**	**5%**	**80%**	**20%**	**100%**
D3	Service ESD (emergency shutdown devices)		✓	✓	
D4	Service and calibrate personal safety systems		✓		
Line E	**PNEUMATIC AND HYDRAULIC SYSTEMS**	**7%**	**60%**	**40%**	**100%**
E1	Examine air supply systems		✓	✓	
E2	Install tubing and fittings		✓	✓	
E3	Install and service pneumatic instruments		✓	✓	
Line F	**ELECTRICAL AND ELECTRONIC SYSTEMS**	**30%**	**40%**	**60%**	**100%**
F1	Examine Electrical Theory		✓		
F2	Apply basic principles of DC electricity		✓	✓	
F3	Apply basic principles of AC electricity		✓	✓	
F5	Apply Boolean logic and principles of digital electronics		✓		
Line G	**FINAL CONTROL ELEMENTS**	**25%**	**40%**	**60%**	**100%**
G1	Service regulators and examine relief valves		✓	✓	
G2	Service, size and install control valves and actuators		✓	✓	
G3	Install and service valve positioners (includes advanced diagnostics)		✓	✓	

		% of Time	% of Time Allocated to: Theory	Practical	Total
Line H	**COMMUNICATIONS, NETWORKING AND SIGNAL TRANSMISSION**	**3%**	**100%**	**0%**	**100%**
H2	Install wiring in accordance with CEC		✓		
Line I	**CONTROL SYSTEMS**	**10%**	**30%**	**70%**	**100%**
I7	Install, configure, maintain and service Programmable Logic Controllers (PLCs)		✓	✓	
	Total Percentage for Industrial Instrument Mechanic Level 1	**100%**			

Training Topics and Suggested Time Allocation

INDUSTRIAL INSTRUMENT MECHANIC – LEVEL 2

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
Line A	**OCCUPATIONAL SKILLS**	**5%**	**60%**	**40%**	**100%**
A3	Use computers and related applications		✓	✓	
A6	Use trade related schematics and drawings		✓	✓	
Line B	**MEASURING AND INDICATING DEVICES**	**45%**	**50%**	**50%**	**100%**
B3	Measure temperature		✓	✓	
B4	Measure level		✓	✓	
B5	Measure density		✓	✓	
B6	Measure weight		✓	✓	
B7	Measure flow (volumetric and mass flow)		✓	✓	
Line E	**PNEUMATIC AND HYDRAULIC SYSTEMS**	**20%**	**40%**	**60%**	**100%**
E4	Install and maintain signal conditioners		✓	✓	
E5	Install and service control devices for hydraulic systems		✓	✓	
Line F	**ELECTRICAL AND ELECTRONIC SYSTEMS**	**20%**	**50%**	**50%**	**100%**
F4	Apply principles of electronics		✓	✓	
Line G	**FINAL CONTROL ELEMENTS**	**10%**	**60%**	**40%**	**100%**
G2	Service, size and install control valves and actuators		✓	✓	
G3	Install and service valve positioners (includes advanced diagnostics)		✓	✓	
	Total Percentage for Industrial Instrument Mechanic Level 2	**100%**			

Training Topics and Suggested Time Allocation

INDUSTRIAL INSTRUMENT MECHANIC – LEVEL 3

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
Line A	**OCCUPATIONAL SKILLS**	**5%**	**80%**	**20%**	**100%**
A2	Organize work and maintain records		✓	✓	
A3	Use computers and related applications		✓	✓	
A6	Use trade related schematics and drawings		✓	✓	
Line B	**MEASURING AND INDICATING DEVICES**	**13%**	**60%**	**40%**	**100%**
B8	Measure consistency and viscosity		✓	✓	
B10	Measure vibration		✓	✓	
B11	Measure speed		✓		
B12	Measure position		✓		
B13	Measure motion		✓		
Line C	**ANALYTICAL INSTRUMENTATION**	**25%**	**40%**	**60%**	**100%**
C2	Measure analytical properties of process liquids		✓	✓	
C3	Measure analytical properties of process solids		✓	✓	
Line E	**PNEUMATIC AND HYDRAULIC SYSTEMS**	**10%**	**50%**	**50%**	**100%**
E3	Install and service pneumatic instruments		✓	✓	
Line G	**FINAL CONTROL ELEMENTS**	**10%**	**50%**	**50%**	**100%**
G4	Install and service variable speed drive (VSD) and variable frequency drive (VFD)		✓	✓	
Line H	**COMMUNICATIONS, NETWORKING AND SIGNAL TRANSMISSION SYSTEMS**	**10%**	**70%**	**30%**	**100%**
H1	Examine communication systems		✓	✓	
H3	Examine communication network structures and components		✓	✓	
H4	Troubleshoot signal transmission systems		✓	✓	

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
Line I	**CONTROL SYSTEMS**	**27%**	**30%**	**70%**	**100%**
I1	Examine fundamental theories of process operation and equipment		✓	✓	
I2	Examine control theory		✓	✓	
I3	Examine process control techniques and strategies		✓	✓	
I4	Implement process control strategies		✓	✓	
I7	Install, configure, maintain and service programmable logic controllers (PLCs)		✓	✓	
I8	Install, configure, maintain and service human machine interface (HMI)		✓	✓	
	Total Percentage for Industrial Instrument Mechanic Level 3	**100%**			

Training Topics and Suggested Time Allocation

INDUSTRIAL INSTRUMENT MECHANIC – LEVEL 4

			% of Time Allocated to:		
		% of Time	Theory	Practical	Total
Line A	**OCCUPATIONAL SKILLS**	**5%**	**40%**	**60%**	**100%**
A3	Use computers and related applications		✓	✓	
A6	Use trade related schematics and drawings		✓	✓	
Line B	**MEASURING AND INDICATING DEVICES**	**10%**	**80%**	**20%**	**100%**
B9	Measure and service environmental monitoring devices		✓	✓	
Line C	**ANALYTICAL INSTRUMENTATION**	**10%**	**50%**	**50%**	**100%**
C1	Measure analytical properties of process gases		✓	✓	
C4	Measure analytical properties of flue gases		✓	✓	
Line D	**SAFETY AND PROCESS MONITORING SYSTEMS**	**15%**	**50%**	**50%**	**100%**
D1	Service and test flame safety systems		✓	✓	
D2	Install and service process cameras		✓		
D3	Service ESD (emergency shutdown devices)		✓		
Line H	**COMMUNICATIONS, NETWORKING AND SIGNAL TRANSMISSION SYSTEMS**	**13%**	**40%**	**60%**	**100%**
H5	Install, configure, maintain and service supervisory control and data acquisition (SCADA) systems		✓	✓	
Line I	**CONTROL SYSTEMS**	**47%**	**40%**	**60%**	**100%**
I1	Examine fundamental theories of process operation and equipment.		✓	✓	
I4	Implement process control strategies		✓	✓	
I5	Install and service stand-alone controllers		✓	✓	
I6	Install, configure, maintain and service Distributed Control Systems (DCS)		✓	✓	
I7	Install, configure, maintain and service Programmable Logic Controllers (PLCs)		✓	✓	
I9	Install and demonstrate knowledge of advanced supervisory control systems		✓	✓	
	Total Percentage for Industrial Instrument Mechanic Level 4	**100%**			

Section 3

PROGRAM CONTENT

Industrial Instrument Mechanic

Level 1

Industrial Instrument Mechanic

Line (GAC): A OCCUPATIONAL SKILLS

Competency: A1 Plan and perform routine trade activities

Objectives:

To be competent in this area, the individual must be able to:

- Perform routine trade activities.

LEARNING TASKS	CONTENT
1. Communicate with others	• Trade terminology • Effective verbal communication skills • Effective written communication skills • Consulting to solve problems
2. Examine types of trade related personal protective equipment	• Head protection o CSA approved hard hat • Eye protection o CSA approved goggles and face shield • Hearing protection o Ear plugs o Ear muffs • Hand protection o Types of gloves and mitts • Clothing o Types of materials suitable to work environment (FR rated) • Foot protection o CSA approved safety boots with suitable soles • Personal Breathing Apparatus
3. Maintain safe work environment	• Safe housekeeping practices • Appropriate recycling and disposal procedures
4. Use and maintain hand and power tools	• Trade specific hand and power tools o (See tools and equipment lists in Appendix)
5. Examine mounting and installation hardware and practices	• Manufacturer instructions • Types of mounting hardware (uni-strut, clamps, u-bolts…) • Location for installation of mounting hardware

LEARNING TASKS	CONTENT
6. Confirm and maintain integrity of test equipment	• Test gauge • Multimeter • Manometer • Dead weight tester • Digital test equipment • Portable personal gas monitors

Achievement Criteria:

Performance	The learner will be evaluated on the ability to: • Use test equipment • Mount and install devices
Conditions	As part of practical lab tasks, given the required tools and materials (see Appendix C: Sample Evaluation Sheet)
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC) A OCCUPATIONAL SKILLS

Competency: A3 Use computers and related applications

Objectives

To be competent in this area, the individual must be able to:

- Configure and program Level 1 instrumentation devices to manufacturers' specifications.

LEARNING TASKS	CONTENT
1. Examines diagnostic and configuration software, hardware and firmware	• Configuration and programming software, hardware and firmware used in Level 1
2. Uses diagnostic and configuration software, hardware and firmware	• Configuration and programming software, hardware and firmware used in Level 1
3. Maintains back-up data and documentation	• Configuration and applicable programming software

Achievement Criteria:

Performance	The learner will be evaluated on the ability to: • Use configuration and programming software, hardware and firmware • Produce back up data and documentation
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): A OCCUPATIONAL SKILLS

Competency: A4 Explain codes, standards and regulations

Objectives:

To be competent in this area, the individual must be able to:

- Access and explain the purpose and applications of standards, codes and regulations.

LEARNING TASKS

1. Navigate WorkSafeBC website to access work-related safety regulations and publications

CONTENT

- http://worksafebc.com
- OHS Regulation
  - Purpose of Regulation
  - General Requirements of OHS
  - Right to refuse unsafe work
  - Government/Employer/ Employee responsibilities
  - Chemical and biological agents
  - Noise, vibration, radiation and temperature
  - Tools machinery and equipment safety
  - Ladders, scaffolds and temporary work platforms
  - Rigging, cranes and hoists
  - Mobile equipment
  - Transportation of workers
  - Traffic control
  - Electrical safety
  - Oil and gas industries
- PDF documents from WorkSafeBC website (publications):
  - Effective Safety and Health Programs
  - Lockout/Tagout
  - Fall Protection
  - Confined Space Hazards
  - Confined Space Entry
  - Working Safely Around Electricity
  - Chlorine Safe Work Practices
  - WHMIS manuals
  - Hazard Symbols Key Booklet
  - Hazard Alerts

LEARNING TASKS	CONTENT
2. Examine safety and certification bodies related to this trade	• Purpose and intent of codes / regulations/standards o WHMIS and use of MSDS o CSA certification standards o ISA documentation o CEC (Canadian Electrical Code) o Boiler and Pressure Vessel Code o CNSC (Canadian Nuclear Safety Commission) o National Energy Board Regulations for Custody Transfer o Oil and Gas Commission Accepted Practices for Measurement o Transportation of Dangerous Goods o BC Mines Act o BC Environmental Regulations • Other related codes and standards, as needed

Line (GAC) A OCCUPATIONAL SKILLS

Competency: A5 Identify electrical hazards and apply safe work practices (includes CEC)

Objectives

To be competent in this area, the individual must be able to:

- Explain standards and safe practices when working with AC and DC electrical circuits and devices.

LEARNING TASKS	CONTENT
1. Examine CEC regulations	• Scope, general rules and applications o Sizing of wire and fuses o Class 1 and Class 2 circuits o Proper installation and grounding of electrical equipment o Area classification o Other sections as needed
2. Examine OHS guide to electrical hazards	• Reference WorkSafeBC Publications – e.g., *Working Safely Around Electricity*

Line (GAC) A OCCUPATIONAL SKILLS

Competency: A6 Use trade related schematics and drawings

Objectives:

To be competent in this area, the individual must be able to:

- Use schematics and drawings related to Level 1 instrumentation.

LEARNING TASKS	CONTENT
1. Examine types of schematics and drawings	• P&ID, SAMA, isometric and orthographic drawings
2. Examine symbols and conventions	• ISA and SAMA symbols
3. Use basic schematics and drawings	• P&ID/P&C drawings

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Use drawings and schematics
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B1 Calibrate and service indicating and recording instruments

Objectives

To be competent in this area, the individual must be able to:

- Calibrate and service specified chart recorders and gauges using principles of links and levers.

LEARNING TASKS	CONTENT
1. Examine types of recording devices	• Chart recorders o Pneumatic o Electronic • Principles of links and levers o Motion multiplication o Angularity o Zero o Span
2. Examine indicating devices	• Displays o Configurable o CRT o LCD/LED o Plasma
3. Calibrate and service indicating devices using principles of zero, span and angularity adjustments as they relate to links and levers	• Gauges o Panel o Field • Accessories o Pigtail siphons o Dampening o Chemical seals • Calculation of head correction • Measuring element and range o Bourdon tube o Helical o Spiral o Bellows o Diaphragm capsule o Slack diaphragm

LEARNING TASKS	CONTENT
	• Applications o Metallurgies o Oil filled o Compound o Combination o Duplex o Differential o Draft o Oxygen service o Refrigeration service
4. Service recording devices	• Identification of measuring element and input measurement scale • Device calibration using principles of zero, span and angularity adjustments as they relate to links and levers • Pen arcing time line • Power supply • Pens • Paper

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate pressure gauges o Draft gauge o Bourdon gauge • Calibrate mechanic, pneumatic and electrical chart recorders
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

LINE (GAC): B MEASURING AND INDICATING DEVICES

Competency: B2 Measure Pressure

Objectives

To be competent in this area, the individual must be able to:

- Configure and calibrate pneumatic, electronic and digital measuring devices to process requirements.

LEARNING TASKS	CONTENT
1. Examine types of pressure	• Absolute • Differential • Gage • Vacuum • Conversion tables o Pressure conversion formulas o Steam tables (relationship between temperature and pressure) • Head correction calculation
2. Examine types of pressure measuring devices	• Pneumatic • Electronic • Digital
3. Examine installation of pressure measuring devices	• Manufacturers' specifications • Selection of device • Air/power supply requirements • Location of device • Isolation of device • Connection of device to process • Connection of device to control system • Sealants and gaskets

LEARNING TASKS	CONTENT
4. Configure / calibrate pressure measuring devices	• Device operation • Primary calibration standards • Manometer types o Well o Raised Well o Dual tube o Incline o U-tube o Slack tube • Manometer fluids o Mercury o Unity oil o Water o Red oil o Meriam #3 o Fluoroscien • Dead weight testers o Pneumatic o Hydraulic • Calibration/configuration parameters • Interpretation of results • Identification of cause/effect of calibration errors • Adjustments to bring device within calibration parameters • Returning device to service after calibration • Document calibration results
5. Maintain device	• Manufacturers' recommended maintenance procedures

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Configure and calibrate pressure measuring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): D SAFETY AND PROCESS MONITORING SYSTEMS

Competency: D3 Service ESD (emergency shutdown devices)

Objectives

To be competent in this area, the individual must be able to:

- Service emergency shutdown devices (ESDs).

LEARNING TASKS	CONTENT
1. Service ESD	• Manipulating process to allow for servicing o Alerting operations o Awareness of impact on process • Testing ESD components • Alarming o Audible alarms o Visible alarms o Response to alarm o Notifications

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Service ESDs
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): D SAFETY AND PROCESS MONITORING SYSTEMS

Competency: D4 Service and calibrate personal safety systems

Objectives

To be competent in this area, the individual must be able to:

- Identify the types of personal safety systems and explain their applications.

LEARNING TASKS	CONTENT
1. Examine personal gas monitors and standard calibration routines	• Portable personal gas monitor (Cl, SO_2, H_2S, O_2 , LEL, CO) • Pull tube (Draeger)
2. Examine radiation safety devices	• Radiation (gamma) survey meter • Personal dosimeter

Line (GAC): E PNEUMATIC AND HYDRAULIC SYSTEMS

Competency: E1 Examine air supply systems

Objectives

To be competent in this area, the individual must be able to:

- Explain the purpose, operation and servicing of air supply systems.

LEARNING TASKS	CONTENT
1. Examine instrument air systems and equipment	• Need for clean, dry air • Air compressors • Air dryers • Air receivers • Air filters
2. Examine air distribution systems	• Mill air • Instrument air • System requirements
3. Use relative humidity to infer dew point	• Chilled mirror • Hygrometer • Hair hygrometer • Sling psychrometer • Digital psychrometer • Bulk polymer resistance sensor • Psychrometric chart
4. Examine the servicing procedures for air supply systems	• Servicing requirements o Traps o Dessicant o Pre and post filters

Achievement Criteria:

Performance	The learner will be evaluated on the ability to: • Measure dew point • Create an instrument air supply drawing from an existing system
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): E PNEUMATIC AND HYDRAULIC SYSTEMS

Competency: E2 Install tubing and fittings

Objectives:

To be competent in this area, the individual must be able to:

- Select, assemble and install tubing and assorted fittings as per drawings provided.

LEARNING TASKS	CONTENT
1. Examine types of tubing and installation procedures	• Plastic • Stainless steel • Copper • Rubber • Process and pressure requirements o Sizes o Pressure and temperature ratings
2. Examine types of fittings and installation procedures	• Types of fittings o Unions o Elbows o Tees o Couplings o Bushings o Reducers o Caps o Plugs o Bulkhead fittings o Others • Tube fittings o Compression o Flared o Hydraulic • Process and pressure requirements o Sizes o Pressure and temperature ratings • Pipe fittings o Ratings
3. Examine tube bending techniques	• Calculating dimensions • Manual tube benders • Hydraulic tube benders

LEARNING TASKS	CONTENT
4. Install tubing and fittings	• Ferrule construction and location • Tightening fittings • Follow P&ID drawings • Select appropriate tubing and fittings

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Identify types of fittings • Bend tubing to a pre-determined pattern
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70% for both tasks

Line (GAC): E PNEUMATIC AND HYDRAULIC SYSTEMS

Competency: E3 Install and Service Pneumatic Instruments

Objectives

To be competent in this area, the individual must be able to:

- Calibrate pneumatic instruments to required specifications.

LEARNING TASKS	CONTENT
1. Examine specifications and hazards of pneumatic equipment	• Compressed air safety • Pneumatic signals (3-15 psi, 6-30 psi, 20-100 kPa) • Required air supplies
2. Examine types of pneumatic equipment	• Transmitters • Converters • Positioners • Controllers • Relays
3. Examine operating principles of pneumatic equipment	• Force balance • Motion balance
4. Examine pneumatic equipment installation procedures	• Selection of equipment o Application o Materials • Location • Set up and adjustments • Isolation of equipment • Repair and replacement methods • Component selections
5. Calibrate pneumatic transmitters	• Force balance calibration procedure • Motion balance calibration procedure • Documentation of calibration results

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate pneumatic equipment
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): F ELECTRICAL AND ELECTRONIC SYSTEMS

Competency: F1 Examine electrical theory

Objectives

To be competent in this area, the individual must be able to:

- Explain principles, sources, types and measures of electrical power.
- Apply related mathematical formulas.

LEARNING TASKS	CONTENT
1. Examine basic principles of electrical theory	• Atomic structure • Conductivity of an element o Conductor o Insulator o Semiconductor • Electrical current o Conventional Theory o Electron Theory
2. Examine sources of AC/DC electrical energy	• Generating electricity o Friction o Temperature differences o Light o Pressure o Chemical reactions o Magnetism • Magnetic Lines of Force • Magnetic induction o AC alternating current o Generated by power plants by magnetic induction o Voltage o Voltage levels o Polarity • DC voltage sources o Fixed polarity – Constant voltage – Fixed direction of flow in a circuit
3. Examine voltage, current and resistance	• Voltage • Amperage • Resistance

LEARNING TASKS	CONTENT
4. Explain Ohm's law	• The relationship between Voltage (E), current (I) and resistance (R) in an electrical circuit • E= I x R

Line (GAC): F ELECTRICAL AND ELECTRONIC SYSTEMS

Competency: F2 Apply basic principles of DC electricity

Objectives

To be competent in this area, the individual must be able to:

- Explain and apply basic principles of DC electricity using DC electrical equipment and instruments.

LEARNING TASKS	CONTENT
1. Examine operation and applications of various batteries	• Lead acid • NiCad • NiMh • Lithium ion
2. Measure electrical current, voltage and resistance	• Analog multimeters • Digital multimeters
3. Calculate currents, voltages and resistance using Ohm's law	• Series circuits • Parallel and combination circuits • Formula E= I x R
4. Define and reference voltage measurement to circuit common	• Difference between ground and circuit common • Multimeter • Oscilloscope and scope meter • Circuit schematic
5. Calculate electrical power in watts	• Apply Watt's Law to define power rating of appliances • Watts = E x I
6. Examine resistors, potentiometers and rheostats	• Differences • Power ratings • Applications • Colour codes
7. Apply appropriate sections of CEC	• Scope, general rules and definitions of the CEC

Achievement Criteria:

Performance — The learner will be evaluated on the ability to:

- Design and build a circuit
- Test for accuracy by calculating and measuring current, voltage and resistance
- Define and reference voltage measurements

Conditions — As part of practical lab tasks, given the required tools and materials

Criteria — Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): F ELECTRICAL AND ELECTRONIC SYSTEMS

Competency: F3 Apply basic principles of AC electricity

Objectives

To be competent in this area, the individual must be able to:

- Explain and apply basic principles of AC electricity using AC circuits.

LEARNING TASKS	CONTENT
1. Define AC electricity	• Generation • Polarity and waveform analysis
2. Examine various types of transformers	• Step up • Step down • Automatic • SOLA • Isolation
3. Examine the use of capacitors and inductors in AC circuits	• Applications • Filtering • Regulating voltage • Power factor correction
4. Size electrical components for various circuits	• Capacitors • Inductors • Resistors • Wire • Fuses
5. Build and test circuits	• Demonstrate use of various AC components in circuits • Measuring techniques and equipment • Sizing components
6. Types of AC circuits	• Class 1 • Class 2 • Section 16 CEC
7. Examine installation procedures for AC equipment	• Wiring methods (Section 12 CEC) • Support • Grounding • Shielding
8. Apply proper circuit connection techniques	• Soldering • Crimping • Printed circuit board repair

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Size electrical components • Build and test AC circuits • Apply proper circuit connection techniques
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): F ELECTRICAL AND ELECTRONIC SYSTEMS

Competency: F5 Apply Boolean logic and principles of digital electronics

Objectives

To be competent in this area, the individual must be able to:

- Explain the principles of digital electronics in logic applications.

LEARNING TASKS	CONTENT
1. Examine principles of digital logic	• Discrete values • Waveforms • Logic levels • Conversions o Digital to analog o Analog to digital o Binary to decimal o Sum of weights o Octal to decimal o Decimal to octal o Binary to octal o Binary to hexadecimal • Logic gate symbols o NOT circuit o Negation and polarity indicators o AND gate o OR gate o NAND gate o NOR gate o XOR gate o XNOR gate
2. Examine digital signal processing	• Analog to digital conversion • Digital to analog conversion • Signal to noise ratio o Analog and digital filters • Signals transformation • Magnitude • Phase • Karnaugh Maps

Line (GAC): G FINAL CONTROL ELEMENTS

Competency: G1 Service regulators and examine relief valves

Objectives

To be competent in this area, the individual must be able to:

- Service regulators.
- Explain the operation of relief valves.

LEARNING TASKS	CONTENT
1. Examine regulators	• Purpose • Pressure drops • Types o Relieving o Non- relieving o Pilot operated • Definitions o Droop o Turndown • Applications o Pressure reducing o Pressure relieving
2. Examine operation and applications of regulators	• Air • Water • Steam • Oil • Gas • Differential
3. Service and maintain regulators	• Components o Diaphragms o Bolts o Springs o Seats o Gaskets • Disassembling o Spring compression • Inspect • Reassemble • Test

LEARNING TASKS	CONTENT
4. Examine relief valves	• Applications o Safety device • Reset differential • Certification and testing

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Service regulators
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): G FINAL CONTROL ELEMENTS

Competency: G2 Service, size and install control valves and actuators

Objectives

To be competent in this area, the individual must be able to:

- Service control valves.
- Install and service actuators.

LEARNING TASKS	CONTENT
1. Examine actuators	• Types o Pneumatic o Hydraulic o Electric • Applications o Fail open o Fail close o Fail last • Actions o Spring return o Double-acting • Components o Diaphragms o Plates o Stem connector (coupling) o Bushings o O-rings o Pistons o Motors o Springs • Required Operating Environment
2. Examine control valves	• Process applications o Metallurgy o Seal/shut off requirements • Flow Characteristics o Quick opening o Linear o Equal percentage

LEARNING TASKS	CONTENT
	• Body Types o Sliding stem – Globe – Bar stock – Pinch valve o Rotary – Butterfly – E-Disc – Segmented ball – Through-bore ball – Restricted trim • Components o Cages o Plugs o Seats o Stems o Packing • Types and applications of valve packing o Teflon o Graphite o Rope
3. Service control valves	• Gaskets • Sealants • Positioning valve in process • Securing valve using appropriate process o Flanged o Screwed o Wafered/flangeless • Isolation of valve from process • Testing procedures o Stroke to ensure proper operation o Leak testing • Possible faults o Leaking packing o Valve passing o Damaged parts o Incorrect travel • Cleaning/lubricating • Repairing/rebuilding

LEARNING TASKS

4. Install and service actuators

CONTENT

- Matching to valve
- Connecting to valve
  - Lifting procedures
- Valve travel
- Bench set
- Verifying operation
  - Correct air supply pressure
- Function testing
- Possible faults
  - Leaking diaphragms
  - Broken springs
  - Damaged/worn O-rings
- Removing/replacing components
- Cleaning/lubricating components
- Assembling/disassembling
  - Spring compression
  - Loading on stem connector
- Returning to service

Achievement Criteria:

Performance	The learner will be evaluated on the ability to: • Service control valves • Remove, service and install actuators on control valves
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): G FINAL CONTROL ELEMENTS

Competency: G3 Install and service valve positioners (includes advanced diagnostics)

Objectives

To be competent in this area, the individual must be able to:

- Install and service valve positioners on final control elements.

LEARNING TASKS

1. Examine valve positioners

CONTENT

- Types
  - Pneumatic
  - Electronic
  - Digital
  - Electro hydraulic
  - Electro mechanical
- Applications
  - Sliding stem/rotary
  - Piston/diaphragm
- Components
  - Levers
  - Nozzles
  - Flappers
  - Relays
- Auxiliaries
  - Locks
  - Boosters
  - Speed controls
- Parameters
- Relation to actuator type/application

LEARNING TASKS

2. Install and service valve positioners

CONTENT

- Mounting
- Connecting to actuator
- Connecting to process control system
- Configuring
  - Set stroke
  - Set pressures
  - Match to actuator
- Calibrating
  - Connecting calibration instruments
  - Interpretation of calibration results
  - Cause/effect of calibration errors
- Component maintenance
  - Remove
  - Replace
  - Repair
  - Clean
- Returning to service

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Install and service valve positioners
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): H COMMUNICATIONS, NETWORKING AND SIGNAL TRANSMISSION SYSTEMS

Competency: H2 Install wiring in accordance with CEC

Objectives

To be competent in this area, the individual must be able to:

- Examine wiring installations in accordance with CEC requirements.

LEARNING TASKS

1. Examine wiring installation requirements

CONTENT

- Materials
- Connections
  - Crimping
  - Terminal blocks
  - Marrettes
  - Soldering
  - Protection (heat shrink, taping etc.)
- Shielding
- Grounding
- Grounding loops
- CEC requirements
- Sizing wire
- Routing of wiring runs
- Stripping wire
- Labeling/colour-coding wire
- Connecting wire

Line (GAC): I CONTROL SYSTEMS

Competency: I7 Install, configure, maintain and service programmable logic controllers (PLCs)

Objectives

To be competent in this area, the individual must be able to:

- Explain the basics of programmable logic controllers (PLCs), given introductory materials on PLCs.

LEARNING TASKS	CONTENT
1. Examine types of PLCs	• Hardware architecture • Control capabilities o Discrete control o Analog control • Compatibility with other process systems • Networks • Protocols
2. Examine PLC languages and symbols	• Structured text • Instruction list • Ladder logic • Function block • Sequential function chart
3. Examine PLC components	• CPU • Memory organization • Input interface • Output interface • Power supply • Programming/monitoring interface • Data table • User program

Level 2

Industrial Instrument Mechanic

Line (GAC): A OCCUPATIONAL SKILLS

Competency: A3 Use computers and related applications

Objectives

To be competent in this area, the individual must be able to:

- Configure and program Level 2 instrumentation devices to manufacturers' specifications given related hardware, software and firmware.

LEARNING TASKS	CONTENT
1. Examines diagnostic and configuration software, hardware and firmware	• Configuration and programming software used in Level 2 o Flow element sizing programs o Temperature and density signal linearization
2. Uses diagnostic and configuration software, hardware and firmware	• Configuration and programming software used in Level 2 o Flow element sizing programs o AGA Mass flow computers

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Perform computerized flow calculations • Program an AGA mass flow computer
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): A OCCUPATIONAL SKILLS

Competency: A6 Use trade related schematics and drawings

Objectives

To be competent in this area, the individual must be able to:

- Use schematics and drawings related to instrumentation.

LEARNING TASKS	CONTENT
1. Examine types of schematics and drawings	• P&ID, SAMA, isometric, orthographic and loop drawings
2. Examine symbols and conventions	• ISA and SAMA symbols
3. Use and modify basic schematics and drawings	• P&ID / P&C / loop drawings

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Use and modify Level 2 drawings and schematics
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B3 Measure temperature

Objectives

To be competent in this area, the individual must be able to:

- Install, calibrate and service temperature measuring devices.

LEARNING TASKS	CONTENT
1. Examine temperature scales	• Fahrenheit • Celsius • Kelvin • Conversions between scales
2. Examine temperature measuring devices and their operation	• Thermometer • Thermocouple o Thermocouple tables • Resistive Thermal Device (RTD) o RTD tables • Thermistor • Filled thermal system • Pyrometer • Semi-conductor mechanical thermal system • Infrared radiation • Fibre Optic
3. Examine temperature calibrating instruments	• Thermometers • Multimeters o Millivolt source o Resistance source • Temperature baths • Dry block calibrators • Thermocouple simulators • Decade box – electronic and analog • Accuracy • Calibration parameters of temperature measuring devices

LEARNING TASKS

4. Installs, calibrates and services temperature measuring devices

CONTENT

- Manufacturers' specifications
- Best Practices for selection/location of measuring device
  - Response time
  - Temperature ranges
  - Resolution
- Thermowell selection and installation
  - Metallurgy
  - Heat transfer
- Thermocouples
  - Grounding
  - Cold junction compensation
  - Types (J, K…T)
  - Extension wires
  - Colour codes (note: North American and European colour codes are different)
    - North American
    - European
- RTDs
  - Alpha and DIN standards
  - 2, 3 and 4 wire
  - 100, 200…1000 ohm
- Device check/calibration
  - Wheatstone bridge
  - Simulators
  - Decade box
- Interpretation of calibration results
- Cause/effect of calibration error
- Device adjustments
- Repairing/replacing device components
- Verification of operation
- Returning device to service
- Documenting calibration

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Assess temperature installations to confirm best practices • Calibrate and service temperature measuring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B4 Measure level

Learning Objectives

To be competent in this area, the individual must be able to:

- Install, calibrate and service level measuring devices.

LEARNING TASKS	CONTENT
1. Examine level measuring devices and their operation	• Point level o Capacitance o Float switches o Tuning fork o Bindicator o Microwave o Ultrasonic o Nuclear • Continuous level o Hydrostatic head o Laser o Ultrasonic o Radar o Sight glass o Bubble pipe o Resistance tape o Magnetic float o Load cell o Displacement o Capacitance • Drum level
2. Examine calibration instruments used on level measuring devices	• Pressure calibrator • Laptop/software • Handheld programmer (configurator)

LEARNING TASKS

3. Install, calibrate and service level measuring devices

CONTENT

- Manufacturers' specifications
- Selection/location of measuring device
  - Process application
  - Process medium
  - Price
  - Best practices
- Verify operation
- Device check/calibration
- Interpretation of calibration results
- Cause/effect of calibration error
- Device adjustments
- Repairing/replacing device components
- Verification of operation
- Returning device to service
- Documenting calibration

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Assess level installations to confirm best practices • Calibrate and service level measuring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B5 Measure density

Objectives

To be competent in this area, the individual must be able to:

- Install, calibrate and service density measuring devices.

LEARNING TASKS	CONTENT
1. Examine density measuring devices and their operation	• Types o Hydrometer o Hydrostatic head o Displacers o Nuclear o Refractometer o Boiling point rise o Coriolis meters • Effect of temperature on density
2. Examine calibration instruments used on density measuring devices	• Pressure calibrator • Laptop/software • Handheld programmer (configurator)
3. Install, calibrate and service density measuring devices	• Manufacturers' specifications • Selection/location of measuring device o Process application o Process medium o Price o Best practices • Verify operation • Device check/calibration • Interpretation of calibration results • Cause/effect of calibration error • Device adjustments • Repairing/replacing device components • Verification of operation • Returning device to service • Documenting calibration • Radiation source regulatory safety test

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Assess density installations to confirm best practices • Calibrate and service density measuring devices • Perform safety tests on a radiation source
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B6 Measure weight

Objectives

To be competent in this area, the individual must be able to:

- Install, calibrate and service weight measuring devices.

LEARNING TASKS	CONTENT
1. Examine weight measuring devices and their operation	• Load cells • Scales • Strain gauges
2. Examine calibration instruments used on weight measuring devices	• Test weights • Calibration chains • Wheatstone bridge • Laptop/software • Handheld programmer (configurator)
3. Install, calibrate and service weight measuring devices	• Manufacturers' specifications • Selection/location of measuring device o Process application o Cost o Best practices • Verify operation • Device check/calibration • Interpretation of calibration results • Cause/effect of calibration error • Device adjustments • Repair/replace device components • Returning device to service • Documenting calibration

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Assess weight installations to confirm best practices • Calibrate and service weight measuring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B7 Measure flow (volumetric and mass flow)

Objectives

To be competent in this area, the individual must be able to:

- Install, calibrate and service flow measuring devices to process requirements.

LEARNING TASKS	CONTENT
1. Examine flow measuring devices and their operation	• Bernoulli's Theorem • Differential pressure o Orifice plate o Flumes/weirs o Annubar o Pitot tube o Target meter o Elbow meter o Venturi o Wedge o Flow nozzle o Multi-variable mass flow o Variable area flow meters • Velocity o Turbine o Vortex o Ultrasonic o Magnetic flow meter • Mass flow o Coriolis o Thermal • Positive displacement meter • Other flow measurement devices
2. Examine calibration instruments used on flow measuring devices	• Pressure calibrators • Flow simulators • Temperature calibrator • Frequency generator • Laptop/software • Handheld programmer (configurator)

LEARNING TASKS

3. Install, calibrate and service flow measuring devices

CONTENT

- Manufacturers' specifications
- Selection/location factors
  - Straight pipe requirements
  - Accuracy requirements
  - Process application
  - Process medium
  - Cost
  - Best practices
- Verify operation
- Device check/calibration
- Interpretation of calibration results
- Cause/effect of calibration error
- Device adjustments
- Repair/replace device components
- Returning device to service
- Documenting calibration

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Assess flow installations to confirm best practices • Calibrate and service flow measuring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): E PNEUMATIC AND HYDRAULIC SYSTEMS

Competency: E4 Install and maintain signal conditioners

Objectives

To be competent in this area, the individual must be able to:

- Calibrate and service signal conditioners to process requirements.

LEARNING TASKS	CONTENT
1. Examine signal conditioners and their operation	• Pneumatic relays o Signal converters o Volume boosters • I/P, P/I transducers • Hardware and software o E.g., high select, function block • Square root extraction • Integrators
2. Calibrate and service signal conditioners	• Manufacturers' specifications for installation o E.g., Moore Industries, Fisher 846, Rosemount

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and service signal conditioners
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): E PNEUMATIC AND HYDRAULIC SYSTEMS

Competency: E5 Install and service control devices for hydraulic systems

Objectives

To be competent in this area, the individual must be able to:

- Explain the types of hydraulic equipment, its specifications and hazards
- Diagnose control devices for different types of hydraulic equipment

LEARNING TASKS	CONTENT
1. Examine hydraulic specifications and hazards	• Contamination o Types o Sources • Fluid cleanliness standards • Filter media o Types o Ratings o Selection o Lifespan o Housing selection • Filter location • Fluid analysis
2. Examine different types of hydraulic equipment	• Types o Pumps o Relays o Regulators • Components o Seals o Spring o Pistons
3. Diagnose control devices for hydraulic systems	• Cleaning o Solvents o Brushes • Connections o To system o Defective • Repair • Valves • Pumps • Sensors

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Diagnose hydraulic control systems
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): F ELECTRICAL AND ELECTRONIC SYSTEMS

Competency: F4 Apply principles of electronics

Objectives

To be competent in this area, the individual must be able to:

- Install and service electronic equipment to manufacturers' specifications.

LEARNING TASKS	CONTENT
1. Examine electronic equipment and its operation	• Analog and digital • Discrete components and their operation o Transistors o Op amps o Diodes o Zener diodes • Power supplies o Half and full wave rectified o Switching o Bridges o Filtering o UPS systems
2. Install and troubleshoot electronic equipment	• Select equipment o Application o Components • Select/install wiring o Current loops o Wiring – 2, 3 and 4 wire transmitters o I/I o Ground loops o Manufacturer's specifications • Connect to system • Adjust settings • Creating and updating loop drawings and documentation

LEARNING TASKS	CONTENT
3. Service electronic equipment	• Isolate equipment • Repair/replacement methods and equipment o Oscilloscope (Scope meter) o Multimeter o Logic probe • Electronic assemblies o Troubleshooting to board level – Power supply – Input conditioning – Signal manipulation – Output circuit – Back plane o Board replacement procedures – Ground strap – Power down and Power Up • Cleaning methods

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Troubleshoot electronic equipment to board level
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): G FINAL CONTROL ELEMENTS

Competency: G2 Service, size and install control valves and actuators

Objectives

To be competent in this area, the individual must be able to:

- Determine the sizing and selection of control valves and actuators.

LEARNING TASKS	CONTENT
1. Examine sizing and selection of actuators	• Size and force required by process conditions
2. Examine sizing and selection of control valves	• Defining C_v • Flow characteristics o Quick opening o Equal percentage o Linear • Process requirements o Medium (Liquid/ gas/steam) o Pressure o Flow o Temperature o Viscosity o TDH (Total Dynamic Head) and NPSH (Net Positive Suction Head) – Correlating pump curve • Flashing/Cavitation • Noise suppression • Sizing for maximum ΔP allowable
3. Select the correct valve type and size for given process applications	• Perform sizing calculations (Liquid/ gas/steam) o Manual (nomograph) • Valve sizing software

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Select correct valve type and size for given process applications
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): G FINAL CONTROL ELEMENTS

Competency: G3 Install and service valve positioners (includes advanced diagnostics)

Objectives

To be competent in this area, the individual must be able to:

- Explain the advanced diagnostics and operational capabilities of Smart Positioners.
- Install, configure and service smart valve positioners.

LEARNING TASKS	CONTENT
1. Examine advanced diagnostics and operational capabilities of Smart positioners	• Determining valve and actuator health o Stiction o Friction o Hysterisis o Duty cycles o Strokes o Travel o Time near closed o Time near open • History • Alarming • Control system interface • Offline diagnostic testing
2. Install and service smart valve positioners	• Mounting • Connecting to actuator • Connecting to process control system • Configuring o Set stroke o Set pressures o Match to actuator o Autotune • Calibrating o Connecting calibration and configuration instruments o Calibraton parameters o Interpretation of calibration results o Cause/effect of calibration errors

LEARNING TASKS

CONTENT

- Component maintenance
  - Remove
  - Replace
  - Repair
  - Clean
- Returning to service

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Install, configure and service smart valve positioners
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Level 3

Industrial Instrument Mechanic

Line (GAC): A OCCUPATIONAL SKILLS

Competency: A2 Organize work and maintain records

Objectives

To be competent in this area, the individual must be able to:

- Estimate labour and material and complete work-related documentation.

LEARNING TASKS	CONTENT
1. Examine and determine standard work requirements	• Materials • Equipment and tools • Personnel • Task planning o Hazard assessment • Work scheduling
2. Examine and apply related skills	• Estimating o Time o Cost o Materials o Manpower needed • Identifying/organizing o Tools o Equipment
3. Examine, maintain and update types of trade related documentation	• Calibration sheets • Data sheets • Work orders • Log entries • Permits • SOP (Standard Operating Procedure) • Management of Change Documentation o Instrument change o Range change o Process change • Maintenance schedules o Preventative o Predictive o Reliability centered • Related software o Spreadsheets o Databases • Word processing

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Estimate labour and material requirements and complete work-related documentation
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): A OCCUPATIONAL SKILLS

Competency: A3 Use computers and related applications

Objectives

To be competent in this area, the individual must be able to:

- Configure and program Level 3 instrumentation devices to manufacturers' specifications given related hardware, software and firmware.

LEARNING TASKS	CONTENT
1. Examines diagnostic and configuration software, hardware and firmware	• Configuration and programming software used in Level 3 o E.g., Autocad, valve sizing software, HMI
2. Uses diagnostic and configuration software, hardware and firmware	• Configuration and programming software used in Level 3 o E.g., Autocad, valve sizing software, HMI
3. Maintains back-up data and documentation	• Configuration and programming software used in Level 3 o E.g., Autocad, valve sizing software, HMI

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Use Level 3 configuration and programming software, hardware and firmware
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

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Line (GAC): A OCCUPATIONAL SKILLS

Competency: A6 Use trade related schematics and drawings

Objectives

To be competent in this area, the individual must be able to:

- Use schematics and drawings related to instrumentation.

LEARNING TASKS	CONTENT
1. Examine types of schematics and drawings	• P&ID, SAMA, isometric and orthographic drawings • Loop drawings
2. Examine symbols and conventions	• P&ID, SAMA, isometric and orthographic drawings • Loop drawings
3. Use and develop schematics and drawings	• P&ID/P&C drawings • Loop drawings

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Use applicable Level 3 drawings and schematics
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B8 Measure consistency and viscosity

Objectives

To be competent in this area, the individual must be able to:

- Confirm installation, calibrate and service consistency and viscosity measuring devices to process requirements.

LEARNING TASKS	CONTENT
1. Examine consistency and viscosity measuring devices and their operation	• Types (analog and Smart) o Optical o Rotary o Blade o Microwave o Nuclear o Viscometer • Factors affecting system performance o Temperature o Flow o Vibration o Pressure o Process considerations
2. Examine instruments and techniques used to calibrate consistency and viscosity measuring devices	• Multimeters • Calibrated weights • Sampling/lab tests
3. Calibrate and service consistency measuring devices	• Manufacturers' specifications • Selection/location factors o Accuracy requirements o Process application o Process medium o Cost o Best practices • Verify operation • Calibration parameters • Device check/calibration • Interpretation of calibration results • Cause/effect of calibration error • Device adjustments • Repair/replace device components • Returning device to service • Documenting calibration

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Assess consistency measuring installations to confirm best practices • Calibrate and service consistency and viscosity measuring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B10 Measure vibration

Objectives

To be competent in this area, the individual must be able to:

- Calibrate and service vibration measuring devices using a vibration monitoring system.

LEARNING TASKS	CONTENT
1. Examine vibration measuring devices	• Probes • Proximitors • Transmitters
2. Examine the installation, calibration and servicing requirements of vibration measuring devices	• Manufacturers' recommended maintenance procedures • Maintenance actions • Identifying cause of calibration errors • Repair and cleaning of device
3. Service vibration monitoring system	• Test and set up vibration monitoring system on operating process equipment

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and service vibration measuring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B11 Measure speed

Objectives

To be competent in this area, the individual must be able to:

- Explain the servicing requirements of speed measuring devices.

LEARNING TASKS	CONTENT
1. Examine relevant laws & principles of physics	• Speed • Velocity
2. Examine speed measuring devices and their applications	• Speed measuring devices o Tachometers o Probes o Proximitors o RPM counters o Strobe lights • Applications o Belt weightometers o Belt slippage o Governors o Radar gun o Interlock o Overspeed trips
3. Examine the installation, calibration and servicing requirements of speed measuring devices	• Manufacturers' recommended maintenance procedures • Maintenance actions • Identifying cause of calibration errors • Repair and cleaning of device

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B12 Measure position

Objectives

To be competent in this area, the individual must be able to:

- Explain the servicing requirements of position measuring devices.

LEARNING TASKS	CONTENT
1. Examine position measuring devices and their applications	• Analog position sensors o LVDT ▪ E.g. Temposonic rods o Proximity switches o Proximity probes o Analog position sensors o Lasers o GPS • Examples of industrial applications o Pulp and paper o Oil and gas o Mining o Food Industry
2. Examine the installation, calibration and servicing requirements of position measuring devices	• Manufacturers' recommended maintenance procedures • Maintenance actions • Identifying cause of calibration errors • Bringing device within calibration parameters • Repair and cleaning of device

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B13 Measure motion

Objectives

To be competent in this area, the individual must be able to:

- Explain the servicing requirements of motion measuring devices.

LEARNING TASKS	CONTENT
1. Examine motion measuring devices and their applications	• Types o Torque switches o Proximity switches o Proximity probes o Analog position sensors o Camera • Applications o Security o Safety o Monitoring rig torque
2. Examine the installation, calibration and servicing requirements of motion measuring devices	• Manufacturers' recommended maintenance procedures • Maintenance actions • Identifying cause of calibration errors • Bringing device within calibration parameters • Repair and cleaning of device

Line (GAC): C ANALYTICAL INSTRUMENTATION

Competency: C2 Measure analytical properties of process liquids

Objectives

To be competent in this area, the individual must be able to:

- Calibrate and service process liquid analyzers to process requirements.

LEARNING TASKS	CONTENT
1. Examine process liquid analyzers	• pH o Measuring electrode o Reference electrode o FET • Conductivity o 2 electrode o 4 electrode o Torroidal • ORP • Specific ion • Dissolved oxygen • Turbidity • Water/effluent treatment o BOD (Biological Oxygen Demand) o COD (Chemical Oxygen Demand) o Silica o Sodium o Residual Chlorine • X-ray Fluorescence
2. Examine process liquid analyzer operational theory and operating parameters	• Non-linear scale • Temperature effects/compensation • Accuracy • Repeatability • Interaction with process • Sources of contamination • Sampling systems • Conditions required

LEARNING TASKS	CONTENT
3. Calibrate and service process liquid analyzers	• Manufacturers' specifications • Selection/location factors o Measurement delays o Chemical mixing o Temperature requirements • Connection to control system or indicator • Configuration of devices • Calibration of devices o Buffering solutions o Calibration standards

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and service process liquid analyzers
Conditions	As part of practical lab tasks, given the required tools, materials and live process equipment
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): C ANALYTICAL INSTRUMENTATION

Competency: C3 Measure analytical properties of process solids

Objectives

To be competent in this area, the individual must be able to:

- Explain the theory and operating parameters of process solids analyzers.

LEARNING TASKS	CONTENT
1. Examine types of process solids analyzers	• Nuclear devices • Assays • Moisture content • X-ray devices • Near infra-red
2. Examine methods used by process solids analyzer	• Online o Material handling considerations o Interface with system • Offline/lab Test o Sample/weigh/dry/weigh o Chemical theory • Standards o ASTM
3. Examine operating parameters of process solids analyzers	• Accuracy • Repeatability • Interaction with process • Sources of contamination • Sampling systems • Conditions required • Method used
4. Examine other process analyzers	• Brightness • Paper sheet scanners • Kappa (K#) analyzers • Sulfidity • Crossbelt analyzers o E.g., Gamma matrix

Line (GAC): E PNEUMATIC AND HYDRAULIC SYSTEMS

Competency: E3 Install and Service Pneumatic Instruments

Objectives

To be competent in this area, the individual must be able to:

- Align pneumatic controllers.

LEARNING TASKS	CONTENT
1. Examine pneumatic controllers	• Force balance • Motion balance
2. Align pneumatic controllers	• Input/output calibration • Temperature and pressure inputs • Indication calibration • Controller alignment and service • Auto/manual transfer stations

Achievement Criteria:

Performance	The learner will be evaluated on the ability to: • Align pneumatic controllers
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): G FINAL CONTROL ELEMENTS

Competency: G4 Install and service variable speed drive (VSD) and variable frequency drive (VFD)

Objectives

To be competent in this area, the individual must be able to:

- Configure and test VSD and VFD.

LEARNING TASKS	CONTENT
1. Examine basic operation of VSD and VFD	• Operation o Tuning parameter identification o Signal isolation DCS/VFD • Control of speed o ECC (Eddy Current Coupling) o Hydraulic speed control o Input signals (digital and analog)
2. Test operation of a VSD/VFD	• Set up and test a VSD/VFD
3. Examine interaction of PID tuning and VSD configuration	• PID control in PLC/DCS with configuration parameters in VSD

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Configure and test VSD and VFD
Conditions	As part of practical lab tasks, given the required tools, materials and load
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry and the learner must achieve a minimum grade of 70%

Line (GAC): H COMMUNICATIONS NETWORKING AND SIGNAL TRANSMISSION SYSTEMS

Competency: H1 Examine communication systems

Objectives

To be competent in this area, the individual must be able to:

- Explain the features and limitations on specified communication protocols.

LEARNING TASKS	CONTENT
1. Examine types of signal transmission systems	• Fibre optics o Armoured cable o Non armoured cable o Multimode/single mode transmission • Wired o Coax o UTP • Wireless o Satellite o Cellular o Blue tooth o RF o IR o IEEE standards
2. Examine features and limitations of communication protocols	• Types of protocols o RS232 o RS422/485 o MODBUS o MODBUS+ o ASi BUS o Device Net o Profibus o Highway Addressable Remote Transducer (HART) o FSK (Frequency Shift Keying) o Foundation Fieldbus o Spread spectrum o Ethernet TCP/IP • Addressing methods and components • Potential sources of interference • Related standards, codes, licenses

Line (GAC): H COMMUNICATIONS NETWORKING AND SIGNAL TRANSMISSION SYSTEMS

Competency: H3 Examine communication network structures and components

Objectives

To be competent in this area, the individual must be able to:

- Explain the basic structures and components of communication networks.

LEARNING TASKS	CONTENT
1. Examine hardware layers	• Network switches (routers) o Configurable o Nonconfigurable o Firewalls o Hubs • Gateways o Protocol interface o Media interface o Network isolation • Hardware topologies o Rapid spanning tree o Self-healing rings o Star
2. Examine network connectors	• Types of connectors o USB o Firewire o 9 pin, 25 pin serial port o RJ45 o RJ11 o M12 o M10 o BNC o Cannon plugs • Resistance and Environmental Standards o IP standards (IP67)

Line (GAC): H COMMUNICATIONS NETWORKING AND SIGNAL TRANSMISSION SYSTEMS

Competency: H4 Troubleshoot signal transmission systems

Objectives

To be competent in this area, the individual must be able to:

- Troubleshoot malfunctioning signal transmission systems to operational requirements.

LEARNING TASKS	CONTENT
1. Troubleshoot wired signal transmission systems	• Current loops o 4-20 mA loops (HART) • Digital buses (at least one of the following): o Foundation Fieldbus o Profibus o Device net • Software configuration o FDT (Field Device Tool) • Performing system diagnostics • Troubleshooting installation problems/ deficiencies o Testing cable o Manipulating process to allow for servicing o Removing/replacing components • Upgrading software and firmware
2. Troubleshoot wireless signal transmission systems	• Signal strength requirements o Batteries • Potential causes of interference • Performing system diagnostics • Troubleshooting installation problems/ deficiencies o Manipulating process to allow for servicing o Removing/replacing components • Upgrading software and firmware • Networks o Line of sight o Spanning tree o Interface to DCS

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Troubleshoot wired and wireless signal transmission systems
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I1 Examine fundamental theories of process operation and equipment

Objectives

To be competent in this area, the individual must be able to:

- Explain the basic operation of common industrial processes.
- Calibrate and tune industrial control loops.
- Diagnose process control problems on a live process.

LEARNING TASKS	CONTENT
1. Examine common industrial processes	• Basic oil & gas field processes o Raw gas processing o Compression o Dehydration • Material handling/quality control o Pulp consistency control process • Separation o Magnets o Screening o Centrifugal o Electrostatic • Concrete plant • Food • Pharmaceuticals • Chemical reaction o pH o Electrolytic o Water and waste water treatment • Evaporation • Flotation
2. Calibrate and tune industrial instrumentation for common industrial processes	• Multiple effect evaporation • Material handling/quality control o Pulp consistency control process • Chemical reaction o E.g. pH o E.g. Conductivity

LEARNING TASKS	CONTENT
3. Operate processes and troubleshoot control problems.	• Multiple effect evaporation • Material handling/quality control o Pulp consistency control process • Chemical reaction o E.g., pH o E.g., Conductivity

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and tune industrial control loops • Diagnose process control problems on a live process
Conditions	As part of practical lab tasks, given the required tools, materials and live process equipment
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I2 Examine control theory

Objectives

To be competent in this area, the individual must be able to:

- Explain basic control theory, actions and operational modes.

LEARNING TASKS	CONTENT
1. Examine basic control theory	• Set point/process variable/ manipulated variable • Relation of output to input • Steady state value and dynamic component • Control loop gains/loop stability
2. Examine control modes	• On/Off control • Differential Gap • Proportional only • Integral only • Proportional plus Integral • PID -Proportional, Integral, Derivative o Reset rate/reset time o Series/parallel o Interactive/non-interactive/ rate on PV
3. Examine controller action	• Direct acting • Reverse acting
4. Examine controller operating modes	• Automatic • Manual • Remote • Local • Supervisory

Line (GAC): I CONTROL SYSTEMS

Competency: I3 Examine process control techniques and strategies

Objectives

To be competent in this area, the individual must be able to:

- Explain control techniques and strategy applications to different processes.

LEARNING TASKS	CONTENT
1. Examine control techniques	• Loop tuning o Zeigler Nicholls o Lambda o Tuning from manual output changes
2. Examine basic control strategies	• Feedback control • Feedforward indexed control • Cascade control • Gap action control • Duplex control • Auto select control

Line (GAC): I CONTROL SYSTEMS

Competency: I4 Implement process control strategies

Objectives

To be competent in this area, the individual must be able to:

- Implement process control strategies.

LEARNING TASKS	CONTENT
1. Implement process control strategies	• Determining required controller action based on process and valve action • Consulting loop diagrams o Overrides o Interlocks o Limits o Select relays • Loop impact on overall process o Permitting procedures o Consulting operators o Alarming • Selecting control strategy • Tuning o 2 mode o 3 mode o Cascade • Implementation on live processes • Upset recovery

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Configure and tune industrial control loops • Diagnose process control problems on a live process
Conditions	As part of practical lab tasks, given the required tools, materials and live process equipment
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): **I** **CONTROL SYSTEMS**

Competency: **I7** **Install, configure, maintain and service programmable logic controllers (PLCs)**

Objectives

To be competent in this area, the individual must be able to:

- Program PLC in ladder logic.
- Troubleshoot various PLCs, given appropriate instructional materials.

LEARNING TASKS	CONTENT
1. Review PLC languages and symbols	• IEC Standard 1131-3 Programming Languages o Instruction List (IL) o Structured Text (ST) o Ladder Diagram (LD) o Function Block Diagram (FBD) o Sequential Function Chart (SFC)
2. Examine and troubleshoot industrial PLC installations	• Hardware • Assembly • Configuration • I/O addressing • Programming o Ladder logic • Data Tables • User Programs
3. Examine and troubleshoot PLC components	• CPU • Memory organization • Input interface • Output interface • Power supply • Programming/monitoring interface • Network communication module
4. Back up and document PLC data for future recovery	• Back up and document programming o Configuration o Settings o Parameters

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Program PLC in ladder logic • Troubleshoot PLC industrial installations and components
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I8 Install, configure, maintain and service human machine interface (HMI)

Objectives

To be competent in this area, the individual must be able to:

- Program HMI software to communicate with a PLC or DCS.

LEARNING TASKS	CONTENT
1. Examine HMI	• Software/hardware design and capability • Compatibility with other process control systems • Communication networks and protocols • Tag descriptors and addressing • Consistency issues in programming • Alarm priorities • Read/write issues • Access/security issues • Communications systems used • Interaction with PLC, DCS
2. Program HMI software	• Program graphical representation of a process • Program HMI software to communicate with a PLC • Build HMI software to interface with PID control and motor control in PLC
3. Back up and document HMI data for future recovery	• Back up and document programming o Configuration • Communication settings

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Program HMI software to communicate with a PLC or DCS
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Level 4

Industrial Instrument Mechanic

Line (GAC): A OCCUPATIONAL SKILLS

Competency: A3 Use computers and related applications

Objectives

To be competent in this area, the individual must be able to:

- Configure and program Level 4 instrumentation devices to manufacturers' specifications given related hardware, software and firmware.

LEARNING TASKS	CONTENT
1. Examines diagnostic and configuration software, hardware and firmware	• Configuration and programming software used in Level 4
2. Uses diagnostic and configuration software, hardware and firmware	• Configuration and programming software used in Level 4

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Use software to configure and program hardware and firmware used in Level 4
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

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Line (GAC): A OCCUPATIONAL SKILLS

Competency: A6 Use trade related schematics and drawings

Objectives

To be competent in this area, the individual must be able to:

- Create basic schematics and drawings.

LEARNING TASKS	CONTENT
1. Create and modify basic drawings	• Electronic drawing o E.g., Autocad • P&ID and SAMA drawings • Loop drawings

Achievement Criteria:

Performance	The learner will be evaluated on the ability to: • Produce a loop sheet drawing
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): B MEASURING AND INDICATING DEVICES

Competency: B9 Measure and service environmental monitoring devices

Objectives

To be competent in this area, the individual must be able to:

- Install, configure and calibrate monitoring devices to process safety requirements.

LEARNING TASKS	CONTENT
1. Examine types of hazardous gases and particulates to be monitored	• Classes and groups of gases • Terms and definitions for hazardous gases o LEL/HEL (Low/High Explosive Limit) o PEL (Personnel Exposure Limit) • Monitored Gases o H2S o CO o Cl2 o SOX o NOX o TRS (Total Reduced Sulphur) • Particulates • Protection o Personnel o Equipment o Environment
2. Examine types of detection equipment for hazardous gases	• Infrared • Catalytic bead • Electro-chemical cell • Lead acid strip • Other technologies
3. Examine operation of monitoring systems	• Acceptable limits • Accuracy limitations • Shut down o Procedures o Actions o Implications

LEARNING TASKS	CONTENT
4. Install, configure and calibrate monitoring devices	• Manufacturers' specifications • Selection/location factors • Connection to control system or indicator • Configuration of devices • Alarming methods • Calibration of devices o Laptop/software o Test gas selection and storage • Documenting calibration

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and service environmental monitoring devices
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): C ANALYTICAL INSTRUMENTATION

Competency: C1 Measure analytical properties of process gases

Objectives

To be competent in this area, the individual must be able to:

- Explain, calibrate and service gas chromatographs.

LEARNING TASKS	CONTENT
1. Examine gas chromatographs	• Gas analysis • Methane, Ethane, Propane, Butane, etc. • Sulfur species
2. Examine gas chromatograph operational theory	• Chromatography • FID (Flame Ionization detector) • PID (Photo Ionization detector) • Thermal Conductivity detector
3. Examine operating parameters of gas chromatographs	• Accuracy • Repeatability • Interaction with process • Sources of contamination • Sampling systems • In situ • Extractive • Conditions required
4. Examine the installation, calibration and servicing of process gas chromatographs	• Manufacturers' specifications • Selection/location factors • Connection to control system or indicator • Configuration of devices • Alarming methods • Calibration of devices • Laptop/software • Test gas selection and storage

LEARNING TASKS

5. Calibrate and service gas chromatographs

CONTENT

- Manufacturers' specifications
- Selection/location factors
- Connection to control system or indicator
- Configuration of devices
- Alarming methods
- Calibration of devices
    - Laptop/software

  Test gas selection and storage

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and service a process gas chromatograph
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): C ANALYTICAL INSTRUMENTATION

Competency: C4 Measure analytical properties of flue gases

Objectives

To be competent in this area, the individual must be able to:

- Explain, calibrate and service flue gas analyzers.

LEARNING TASKS	CONTENT
1. Examine flue gas analyzers	• Online flue gas analyzers o Excess oxygen o CO o Particulate/opacity o TRS o NOX o SOX • Lab tests o Orsat o Fyrite
2. Examine flue gas analyzer operational theory	• Thermo-paramagnetic • Zirconium oxide • Catalytic combustibles detector • Infrared laser
3. Examine operating parameters of flue gas analyzers	• Accuracy • Repeatability • Interaction with process • Sources of contamination • Sampling systems o In situ o Convective o Close-coupled extractive o Extractive
4. Calibrate and service process flue gas analyzers	• Manufacturers' specifications • Selection/location factors • Connection to control system or indicator • Configuration of devices • Alarming methods • Calibration of devices o Laptop/software • Test gas selection and storage

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and service flue gas analyzers
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): D SAFETY AND PROCESS MONITORING SYSTEMS

Competency: D1 Service and test flame safety systems

Objectives

To be competent in this area, the individual must be able to:

- Troubleshoot flame detection equipment.
- Service flame safety systems.

LEARNING TASKS	CONTENT
1. Examine flame detection equipment	• Ultraviolet • Infrared • Magnetic • Rate of rise • Heat sensors (thermopile) • Ionic
2. Examine operation of flame safety systems	• Acceptable limits o BC Safety Authority regulatory requirements • Accuracy • Shut down o Procedures o Actions o Implications • Applications o BMS o Flare stacks
3. Troubleshoot flame detection equipment	• Manufacturers' specifications and recommendations • Selecting required equipment • Connecting to process/indicator • Configuring • Calibrating • Alarming

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Troubleshoot flame detection equipment • Troubleshoot Burner Management Systems (BMS)
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): D SAFETY AND PROCESS MONITORING SYSTEMS

Competency: D2 Install and service process cameras

Objectives

To be competent in this area, the individual must be able to:

- Explain the operation of process cameras and their applications.

LEARNING TASKS	CONTENT
1. Examine process camera applications	• Leak monitoring • Fire monitoring • Intruder alert • Remote monitoring o Process control o Quality control • Safety
2. Examine the operation of process cameras	• Analog and Digital • Manufacturers' specifications and recommendations • Selecting required equipment • Connecting to process/indicator • Configuring • Calibrating • Alarming

Line (GAC): D SAFETY AND PROCESS MONITORING SYSTEMS

Competency: D3 Service ESD (emergency shutdown devices)

Objectives

To be competent in this area, the individual must be able to:

- Explain the types of Emergency Shutdown Devices (ESD), their purposes and testing procedures.

LEARNING TASKS	CONTENT
1. Examine types of ESD control systems	• Levels of shutdown o Equipment shutdown o Area shutdown o Total/Plant shutdown • Types of ESD o Electric o Pneumatic o Hydraulic o Mechanical
2. Examine purposes of different types of ESD	• Personnel protection • Environmental protection • Equipment protection
3. Examine ESD testing procedures	• Partial Stroke test • Time test • Valve integrity • Interlock checks (system shut down check)

Line (GAC): H COMMUNICATIONS, NETWORKING AND SIGNAL TRANSMISSION SYSTEMS

Competency: H5 Install, Configure, Maintain and Service Supervisory Control and Data Acquisition (SCADA) systems

Objectives

To be competent in this area, the individual must be able to:

- Explain SCADA protocols, configurations, equipment and servers.
- Program and service SCADA systems.

LEARNING TASKS	CONTENT
1. Examine types of SCADA protocols and configurations	• Applications o Custody transfer – AGA/API calculations • Online history • Remote equipment operation • Time synchronization and time stamping • Network layout o Protocols o Host o Field • Addressing methods • Configuration licensing
2. Examine types of SCADA equipment and servers for data acquisition and storage	• Radio Telemetry Units (RTU) • Wireless communications systems o Cellular o Satellite o Radio
3. Service SCADA systems	• Manipulating process to allow for servicing • Alerting operations • Transferring from automatic to manual • Awareness of impact on process • Maintaining on site software/firmware revisions and data backups • Performing system diagnostics • Testing SCADA components • Maintaining host integrity • Installation problems and deficiencies • Develop logic strategies

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Program and service SCADA systems
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I1 Examine fundamental theories of process operation and equipment

Objectives

To be competent in this area, the individual must be able to:

- Explain the operation of common industrial processes using advanced control strategies.
- Calibrate and tune industrial control loops.
- Diagnose process control problems on a live process.

LEARNING TASKS	CONTENT
1. Examine industrial instrumentation for advanced control systems with industrial processes	• Steam generation (boilers) • Batch process o Digester • Distillation o Fractionation o Binary tower o Cryogenic o LNG
2. Calibrate and tune industrial instrumentation for advanced control systems with industrial processes	• Steam generation (boilers) • Batch process o Digester • Distillation o Fractionation • Binary tower
3. Operate processes and troubleshoot advanced control systems	• Steam generation (boilers) • Batch process o Digester • Distillation o Fractionation • Binary tower

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Calibrate and tune industrial control loops • Diagnose process control problems on a live process
Conditions	As part of practical lab tasks, given the required tools, materials and live process equipment
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I4 Implement process control strategies

Objectives

To be competent in this area, the individual must be able to:

- Implement advanced process control strategies.

LEARNING TASKS	CONTENT
1. Implement and tune process control strategies	• Determining required controller action based on process and valve action • Consulting loop diagrams o Overrides o Interlocks o Limits o Select relays • Loop impact on overall process o Permitting procedures o Consulting operators o Alarming • Selecting control strategy o Cascade control o Feed forward index control o Feed forward control o Constant Ratio control • Implementation on live processes • Upset recovery

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Implement and tune advanced process control strategies
Conditions	As part of practical lab tasks, given the required tools, materials and live process equipment
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I5 Install and service stand-alone controllers

Objectives

To be competent in this area, the individual must be able to:

- Install, configure and service stand-alone controllers for various control strategies.

LEARNING TASKS	CONTENT
1. Examine stand-alone controllers	• Microprocessor o Single loop o Cascade loop • I/O o HART o Electronic • Installation requirements o Ambient temperature o Area classification
2. Configure stand-alone controller	• Maintenance • Configuration o Techniques o Tools – Hand held programmers – Software

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Configure stand alone controllers for various control strategies
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I6 Install, configure, maintain and service Distributed Control Systems (DCS)

Objectives

To be competent in this area, the individual must be able to:

- Configure DCS equipment.

LEARNING TASKS	CONTENT
1. Examine DCS	• DCS o E.g., Emerson, Schnieder, Honeywell • System configuration • LAN communication protocols • Hardware components • Configuration software • Troubleshooting • Operator console and diagnostic tools o Analog and discrete input and output signals o Tunable parameters in software blocks • SIS systems
2. Configure DCS	• Build and troubleshoot a cascade control system (including operator interface graphics) • Configure and troubleshoot analog inputs, analog outputs, control loops and pump stop/start

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Configure and troubleshoot a DCS
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I7 Install, configure, maintain and service programmable logic controllers (PLCs)

Objectives

To be competent in this area, the individual must be able to:

- Configure and troubleshoot PLCs.

LEARNING TASKS	CONTENT
1. Configure PLC	• Configuration of analog control strategies using a minimum of two of the following IEC 1131 standard protocols: o FBD o ST o IS o SFC • Software interface • Operation • Troubleshooting • SIS systems
2. Back up and document PLC data for future recovery	• Back up and document programming o Configuration o Settings o Parameters

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Configure a PLC
Conditions	As part of practical lab tasks, given the required tools and materials
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Line (GAC): I CONTROL SYSTEMS

Competency: I9 Install and demonstrate knowledge of advanced supervisory control systems

Objectives

To be competent in this area, the individual must be able to:

- Explain and demonstrate process optimization for an advanced supervisory control system.

LEARNING TASKS	CONTENT
1. Examine batch process control	• Operation and application of Batch Process Control
2. Examine Batch Process Control and compare to other control applications and strategies	• Continuous • Discontinuous
3. Examine batch process control software in use	• DCS and PLC • Applications (examples) o Batch pulp digester process o Concrete plant o Oil pipeline transmission o Chemical industry o Food plant
4. Examine and tune boiler control systems	• Combustion control systems o Parallel open/closed loop o Cross limited – Application of excess oxygen trim control • Plant master VS boiler master controls • Application of feedforward control indexing to plant master pressure controller • Steam temperature attemporator • Conventional desuperheater control • 2,3 and 5 element drum level control • Balanced draft furnace pressure control o FD and ID fans
5. Examine other advanced supervisory control systems	• Predictive control techniques o Smith Predictors o MPC
6. Examine and diagnose safety instrumented systems (SIS)	• SIL levels • Voting structures • Documentation

Achievement Criteria

Performance	The learner will be evaluated on the ability to: • Tune boiler control systems
Conditions	As part of practical lab tasks, given the required instrumentation, tools, materials and live process equipment
Criteria	Tasks must be performed within specifications, safety standards and time frames acceptable to industry, and the learner must achieve a minimum grade of 70%

Section 4

TRAINING PROVIDER STANDARDS

Instructor Requirements

Occupation Qualification

The instructor must possess:

- Red Seal Qualification as an Instrumentation and Control Technician (Industrial Instrument Mechanic)

Work Experience

A minimum of 5-years' experience working in the industry as a Journeyperson.

Instructional Experience and Education

It is preferred that the instructor also possesses one of the following:

- Instructors Certificate (minimum 30 hr course)
- Registered in an Instructor's Diploma Program (to be completed within a five year period)
- Bachelor's or Master's degree in Education
- Power Engineering Certificate (4th Class or higher)
- Red Seal qualification as an Industrial or Contruction Electrician

Facility Requirements

General Areas

- Cleaning supplies
- Compliance with all local and national fire code and occupational safety requirements
- Adequate lighting
- Heating/air conditioning for comfort all year round
- 120 volt AC

Classroom Area (General Area requirements plus the following)

- Comfortable seating and tables suitable for training, teaching, and lecturing
- Lighting controls to allow easy visibility of projection screen while also allowing students to take notes
- Windows must have shades or blinds to adjust sunlight
- Heating/air conditioning for comfort all year round with room-specific control
- Acoustics in the room must allow audibility of the instructor
- White marking board with pens and eraser
- Projection screen or projection area at front of classroom
- Document camera and/or multi-media projector

Shop Area (General Area requirements plus the following)

- Lifting devices
    - E.g., overhead cranes
- Workbenches with 6” vices
- Instrument air supply

Lab Requirements

All Levels

- Communication and Signal transmission instrumentation and final control equipment
- Measurement recorders and indicators including motion, speed, vibration, position, mass flow, pH, temperature, pressure, weight, level
- Multiple computer stations with interfacing options
- Resource computer with internet access
- Instrument air supply

Level 1

- Shoebox Programmable Logic Controllers (e.g., Omron, Westinghouse, Schneider)

Level 2

- Shoebox Programmable Logic Controllers (e.g., Omron, Westinghouse, Schneider)
- Access to a radiation source that may be used for level or density measurement

Level 3

- Fully operational, representative process-equipment with supporting instrumentation and control equipment (e.g., distillation column, evaporator and power boiler, pulp stock digester, mineralization)
- Installed control system (e.g., Fisher Delta V)
- Stand alone controllers, pneumatic
- Software-loadable Programmable Logic Controllers (e.g., Schneider, GE/Fanuc, AB)
- Distributed Control Systems (e.g., Schneider Foxboro IA, Delta V, ABB)
- Access to a radiation source that may be used for level or density measurement

Level 4

- Fully operational, representative process-equipment with supporting instrumentation and control equipment (e.g., distillation column, evaporator and power boiler, pulp stock digester, mineralization)
- Installed control system (e.g., Fisher Delta V)
- Stand alone controllers, electronic (e.g., F&P MC-5000)
- PC-based advanced control software (e.g., Brainwave)
- Software-loadable Programmable Logic Controllers (e.g., Schneider, GE/Fanuc, AB)
- Distributed Control Systems (e.g., Schneider Foxboro IA, Delta V, ABB)
- SCADA systems (e.g., Bristol, Fisher, Schneider)
- Access to a radiation source that may be used for level or density measurement

Student Facilities

- Adequate lunch room as per WorkSafeBC requirements
- Adequate washroom facilities as per WorkSafeBC requirements
- Personal storage lockers

Instructor's Office Space

- Private seating space sufficient for 3 people (separate from training space)

Other

- Not applicable

Tools and Equipment

Shop Equipment

Power Tools

Required

- Air compressor
- Drill press
- Grinders
- Heat gun
- Portable electric drill
- Pressure and vacuum pumps
- Soldering iron with appropriate ventilation

Recommended

- Cutoff saw
- High pressure grease gun
- Hydraulic press
- Impact wrench
- Pipe threader
- Powder actuated tools (hilti, ramset, etc.)
- Pneumatic tools

Electronic Tools and Test Equipment

Required

- Amp probe
- Analog multimeter
- Flue gas analyzers (complete with Ringelmann chart)
- Gas chromatograph
- Barometer
- Bridges
- Calibrated oven
- Capacitance simulator
- Current calibrator
- Data logger
- Deadweight tester (hydraulic and pneumatic)
- Decade resistance box
- Deflectional-type strain indicator
- Dew point tester
- Digital multimeter
- Dry block calibrator
- Electromagnetic flowmeter
- Electrostatic voltmeter
- Ethernet network kit
- Frequency counter
- Frequency generator
- Gauge blocks
- Hand held programmer (configurator)
- Millivoltmeter calibrator
- Modem
- Null balance strain indicator
- Optical pyrometer
- Oscilloscope
- pH simulator/buffers
- Pneumatic test stand
- Portable sound level meter
- Portable voltage tester
- Potentiometer
- Power supplies
- Pressure/vacuum calibrator
- Printers
- Protocol analyzer
- Radiation meter
- Regulator
- Rpm tester/tachometer
- Rtd/thermocouple calibrator
- Signal generator
- Signal analyzer
- Sling psychrometer
- Software
- Tachometer generator

- Hand held pyrometer
- Hydrometer
- Infrared thermometer
- Lab scales
- Label maker
- Laptop computer
- Logic testers
- Loop calibrator/simulator
- Manometer (well and incline)
- Temperature bath
- Test gases
- Test gauges (pressure, vacuum)
- Thermal meter
- Thermometer
- Variable transformer
- Vibration table (wobbulator)
- Wrist ground strap

Recommended

- Eddy current tachometer
- Laser strength meter
- Microwave leakage meter
- Stroboscope
- Wireless signal strength tester

Shop (Facility) Tools

Required Hand Tools

- Calipers (assorted)
- Crowbar
- Diagonal cutter
- Drill bits
- Easy-out (extractor)
- Portable vices
- Flaring tool
- Flashlight
- Fuse puller
- Gauge pointer puller
- Gasket cutter
- Grease gun
- Hammers (assorted)
- Hand files (assorted)
- Hex keys (Imperial/Metric)
- Knockout punch
- Level
- Magnet
- Measuring tape
- Micrometers (assorted)
- Mirrors
- Heat shrink labeller
- Nut drivers (assorted)
- Packing puller
- Pinch bar
- Pipe threader
- Pliers (assorted)
- Punches (assorted)
- Reamer
- Rubber mallet
- Saws (assorted)
- Screw starter
- Screwdrivers (assorted)
- Scriber
- Sockets (imperial/metric)
- Square
- Steel rule
- Strap wrench
- Tap and die set
- Pipe wrench
- Torque wrench
- Tube benders
- Tube cutter
- Tweezer
- Wire brushes
- Wire crimpers
- Wire cutter
- Wire labeller
- Wire stripper
- Wrenches
- Toolbox

Personal Protective Equipment and Safety Equipment (provided by TP)

Required

- Ear protection (muffs and plugs)
- Eye wash bottle
- Face shield
- Fire extinguisher
- First aid kit
- Gloves
- Grounding mat
- Hard hat
- Mask
- Personal dosimeter
- Personal monitor (gas)
- Radiation survey meter
- Respirators
- Rubber safety boots
- Rubber suits
- Safety glasses
- Safety harnesses with lanyard
- Splash goggles

Recommended

- Apron
- Cap lamp
- Personal locks for lockout
- Scissor clamps for lockout

Specialty Tools (required)

- Lifting and rigging equipment
- Ventilated fume hood

Student Tools (supplied by student)

NOTE: check with training provider for student equipment and tools

Required

- Steel-toed boots

Recommended

- Coveralls
- Fluke 789 process calibrator or equivalent
- Powered breadboard with lead kit (e.g., MB-800 project board)

Reference Materials

Introduction

The amount of technical data and the rate of technological innovation confronting workers in this trade are extremely high.[1] Manufacturers who formerly maintained in-house publishing operations (due to the sheer quantity of specialized technical information associated with their products) have now moved to providing technical information in on-line and/or CD formats, in order to reduce costs associated with frequent revisions and updates to technical materials. The implications for the development of trade training materials are clear: there is potential for rapid redundancy of information and a need for constant Subject Matter Expert-led evaluation of curriculum.

As well, there are almost infinite variations in the technologies of industrial instrumentation. The focus of BC IIM training is on the technologies *most prevalent* in the industries of this province. This approach has served industry well. Apprentices are prepared to work with the most current BC technologies and also capable of dealing with the older systems sometimes encountered in the oil and gas fields. Training materials should support a "BC first" focus, while enabling apprentices to successfully challenge the IP exam for this trade.

MATERIALS IN PRINT

- *Canadian Electrical Code, Part 1*, most current edition.
- McMillan, Gregory K. **Advanced temperature measurement and control**
- Vickers, Incorporated Training Center **Closed loop electrohydraulic systems manual**
- Warren, John E. **Control instrument mechanisms**
- Coggan, Donald A. **Fundamentals of industrial control**
- Parr, E.A. **Hydraulics and Pneumatics**
- Eaton Corporation **Industrial Hydraulics manual**
- Eaton Corporation **Industrial Hydraulics answer book**
- Liptak, Bela G. **Instrument Engineers Handbook. Process Management and Analysis**
- Liptak, Bela G. **Instrument Engineers Handbook. Process Control and Optimization**
- Liptak, Bela G. **Instrument Engineers Handbook. Process Software and Digital Networks**
- Eaton Fluid Power Training **Introduction to Hydraulics Technology**
- Nyce, David S. **Linear Position Sensors**
- Park, John **Practical Data Communication for Instrumentation and Control**
- Mackay, Steve **Practical Industrial Data Networks**
- Terrel, David L. **Fundamentals of Electronics DC/AC Circuits**
- Cooke and Adams **Basic Math for Electronics**
- Ptec **Instrumentation**
- Kirk, Franklin & Philip, Weeder, Thomas A. **Instrumentation**
- Murrill, Paul W. **Fundamentals of Process Control Theory**
- Buchla, David **Experiments of Digital Fundamentals**
- Floyd, Thomas L. **Principals of Electric Circuits**

[1] When asked about the technical information resources they use in their trade, about 2/3 of the SME's, identified the "Help Menus" of the products they configure and install as a significant source of up-to-date technical information.

- Bartlet, Terry — **Instrumentation and Process Control**
- Patrick, Dale R. & Steven R. — **Pneumatic Instrumentation**
- Faulk, Sutko — **Industrial Instrumentation**
- Thomson, Delmar Learning — **Fundamentals of Instrumentation**
- Price, Winston T. & Miller, Merlin — **Elements of Data Processing Math**
- Alerich, Walter N. & Keljik, Jeff — **Electricity 3**
- Considine, Douglas M. — **Process Industrial Instrumentation and Control Hand Book**
- Anderson, Norman A. — **Instruments for Process Measurement and Control**
- Bell, David A. — **Fundamentals of Electric Circuits**
- Rease, Dudley A. — **Basic Fluid Power**
- Skoog, Douglas A. & West, Donald M. — **Fundamentals of Analytical Chemistry**
- Shortley and Williams — **Elements of Physics**
- Wildi, Theodore — **Electrical Machines, Drives and Power Systems**
- Johnston, Curtis D. — **Process Control Instrument Technologies**
- Steingress, Frederick M. — **Low Pressure Boilers**
- Heath, Macnaughton and Martindale — **Fundamentals of Physics**
- Kuphaldt, Tony R. — **Lessons in Industrial Instrumentation**
- Floyd, Thomas L. — **Digital Fundamentals, 8th edition. Prentice-Hall, 2005.**
- Spitzer, David W. — **Industrial Flow Measurement, 3rd edition. Instrument Society Of America, 2005.**
- Trevathan, Vernon L. — **Ed. A Guide to the Automation Body of Knowledge, 2nd edition. Instrument Society of America, 2006.**
- Eren, Halit — **Wireless communication systems/ Design and construction; CRC Press, 2006. 297 pages ISBN 0849336740**
- Macdonald, Dave — **Practical Industrial Safety, Risk Assessment and Shutdown Systems, 2003.**
  Paperback, 384 pages, publication date: NOV-2003
  ISBN-13: 978-0-7506-5804-1
  ISBN-10: 0-7506-5804-5

ONLINE RESOURCES

(AS OF JULY 2014)

- www.abb.com ABB
- www.boschrexroth.ca Bosch Rexroth Canada is the Canadian partner of Bosch Rexroth, an international company specializing in “Drive and Control.” Some technical information on hydraulics, including course outlines for introduction and maintenance.
- www.control.com “Control.com,” an online global community of automation professionals. Webpage includes a forum for questions, list of topic threads, opportunity for exchange of ideas and information with other instrumentation professionals.
- www.controlglobal.com/whitepapers/
- http://www.controlsweekly.com – Controls Weekly Review – weekly reviews of manufactured systems used in process control; archive; topics list. Information updated weekly.
- www.cpecn.com/
- www.croftinst.com/home.htm Croft Instrument Systems – process instrument designers, suppliers and manufacturers. Process Solids: (see: “suspended solids” and “standard consistency” for technical/product notes).
- www.cvs-controls.com CVS Controls is a manufacturer and supplier of products for the process control industry. Select “literature”: free instruction manuals available.
- www.cyberlaboratory.com/ -- Information on density.
- www.documentation.emersonprocess.com/ Click on “Emerson Process Management Documentation Library” for free downloads, including a 297 page Control Valve handbook.
- www.emersonprocess.com Emerson.
- www.emersonprocess.com/fisher Fisher.
- www.emersonprocess.com/university - PlantWeb University has 11 courses (free download when registered – no cost to register) on Safety Instrumented Systems (SIS) and 21 courses on wireless technologies.
- www.enmet.com Enmet Corporation. Manufactures gas and vapor detectors, stationary and portable.
- www.fisherregulators.com (requires registration to access technology literature).
- www.flowcontrolnetwork.com
- www.foxboro.com Foxboro
- www.galvanic.com Galvanic Applied Sciences Ltd. (see “suspended solids” under the “liquid measurement” heading for product notes).
- www.gongol.net DJ Gongol and Associates, manufacturers of range of process-control related equipment. Select “Instruments” – scroll to “toxic gas detections”, see specifications for portable and hand held equipment.
- www.graceindustries.com Grace Industries manufactures industrial safety products. Information on lone worker security systems (click “industrial Safety products”).
- www.honeywell.com Honeywell.
- www.iceweb.com.au/Technical/LevelTechnologies.html
- www.invensys.com Invensys.
- www.isa.org The Instrumentation, Systems and Automation Society.
- www.joliettech.com Joliet Technologies, producer of variable speed drive systems and controls. Product material has good information on VSD and VFD.
- www.metsoautomation.com Metso.

- http://www.modelingandcontrol.com/ Modeling and Control: the Dynamic World of Process Control is a blog written by two men with a “broad range of experience in the design and commissioning of batch and continuous process control systems and the development and application of process simulation for operator training and control study.” They write with the intent that readers will find the information posted interesting and helpful in work situations.
- www.nfpa.org/codes-and-standards/document-information-pages NFPA Codes and Standards e.g. NFPA 85 – Boiler and Combustion Systems Hazards Code.
- http://www.ca.endress.com/en Endress + Hauser.
- http://www.blrbac.org/ Black Liquor Recovery Boiler Advisory Committee.
- http://www.multimediahrd.com/ Multi media offers DVD and video materials on 10 topics related to hydraulics training. Click on “DVD and video” on webpage sidebar, scroll down to “technical” on new page shown, select “hydraulics” to view topics covered.
- www.omega.com – Information on basic process measurements like flow, temperature, pressure, pH, conductivity, level, etc.
- www.ohsonline.com --National US website on employment safety issues; use Search button to get information on personal gas detectors.
- www.processingtalk.com/guides/ News and information site for Process Engineers, updated daily. Select “Emergency Shutdown” from list of common terms – or browse through for information on other topics.
- www.raesystems.com Rae Systems. See technical and application notes for information on hand held and portable sensors (personal safety systems).
- www.scadalink.com Bentek Systems. See Tech notes for information on wireless SCADA systems.
- www2.sea.siemens.com/Products/Process-Instrumentation/Support/PI-User-Manuals --advanced control strategies.
- www.smar.com/PDFs/Catalogues/FBTUTCE.pdf -- Foundation Fieldbus information. www.smar.com/PDFs/Catalogues/HARTTUTCE.PDF -- a good tutorial on HART communication.
- www.spitzerandboyes.com
- www.vegacontrols.co.uk/vega_downloads_open.htm --Radar and ultrasonic level measurements.
- www.worksafebc.com -- WorkSafeBC’s webpage – view the provincial OHS regulation, which explains employer/employee responsibilities, get access to WorkSafeBC publications on specific issues (young worker safety, accident reports…etc.)
- www.yokogawa.com Yokogawa.
- www.zoneni.com National Instruments – see the NI developer zone.
- http://www.itabc.ca/program/instrumentation-and-control-technician-industrial-instrument-mechanic http://nuclearsafety.gc.ca/eng/ Canadian Nuclear Safety Commission.
- http://www.nist.gov/ - National Institute of Standards and Technology.

Appendices

Appendix A

Assessment Guidelines

Grading Sheet: Subject Competency and Weightings

PROGRAM: IN-SCHOOL TRAINING: ITA DIRECT ACCESS CODE:	INDUSTRIAL INSTRUMENT MECHANIC LEVEL 1		
LINE	**SUBJECT COMPETENCIES**	**THEORY WEIGHTING**	**PRACTICAL WEIGHTING**
A	Applies Occupational Skills	9%	2%
B	Installs and Maintains Measuring and Indicating Devices	13%	17%
D	Installs and Maintains Safety and Process Monitoring Systems	8%	2%
E	Installs and Maintains Pneumatic and Hydraulic Systems	9%	5%
F	Installs and Maintains Electrical and Electronic Systems	26%	33%
G	Installs and Maintains Final Control Elements	21%	28%
H	Installs and Maintains Communications, Networking and Signal Transmission	7%	0%
I	Installs and Maintains Control Systems	7%	13%
	Total	**100%**	**100%**
In-school theory / practical subject competency weighting		70%	30%
Final in-school percentage score		IN-SCHOOL %	

In-school Percentage Score Combined theory and practical subject competency multiplied by	80%
Standard Level Exam Percentage Score The exam score is multiplied by	20%
Final Percentage Score	FINAL%

PROGRAM: IN-SCHOOL TRAINING: ITA DIRECT ACCESS CODE:	INDUSTRIAL INSTRUMENT MECHANIC LEVEL 2		
LINE	**SUBJECT COMPETENCIES**	**THEORY WEIGHTING**	**PRACTICAL WEIGHTING**
A	Applies Occupational Skills	6%	4%
B	Installs and Maintains Measuring and Indicating Devices	46%	49%
E	Installs and Maintains Pneumatic and Hydraulic Systems	11%	10%
F	Installs and Maintains Electrical and Electronic Systems	25%	29%
G	Installs and Maintains Final Control Elements	12%	8%
	Total	**100%**	**100%**
In-school theory / practical subject competency weighting		70%	30%
Final in-school percentage score		IN-SCHOOL %	

In-school Percentage Score Combined theory and practical subject competency multiplied by	80%
Standard Level Exam Percentage Score The exam score is multiplied by	20%
Final Percentage Score	FINAL%

PROGRAM: IN-SCHOOL TRAINING: ITA DIRECT ACCESS CODE:	INDUSTRIAL INSTRUMENT MECHANIC LEVEL 3		
LINE	**SUBJECT COMPETENCIES**	**THEORY WEIGHTING**	**PRACTICAL WEIGHTING**
A	Applies Occupational Skills	5%	2%
B	Installs and Maintains Measuring and Indicating Devices	21%	10%
C	Installs and Maintains Analytical Instrumentation	21%	28%
E	Installs and Maintains Pneumatic and Hydraulic Systems	11%	9%
G	Installs and Maintains Final Control Elements	6%	9%
H	Installs and Maintains Communications, Networking and Signal Transmission Systems	10%	6%
I	Installs and Maintains Controls Systems	26%	36%
	Total	**100%**	**100%**
In-school theory / practical subject competency weighting		70%	30%
Final in-school percentage score		IN-SCHOOL %	

In-school Percentage Score Combined theory and practical subject competency multiplied by	80%
Standard Level Exam Percentage Score The exam score is multiplied by	20%
Final Percentage Score	FINAL%

PROGRAM: IN-SCHOOL TRAINING: ITA DIRECT ACCESS CODE:	INDUSTRIAL INSTRUMENT MECHANIC LEVEL 4 / FINAL LEVEL		
LINE	**SUBJECT COMPETENCIES**	**THEORY WEIGHTING**	**PRACTICAL WEIGHTING**
A	Applies Occupational Skills	4%	6%
B	Installs and Maintains Measuring and Indicating Devices	12%	4%
C	Installs and Maintains Analytical Instrumentation	16%	9%
D	Installs and Maintains Safety and Process Monitoring Systems	16%	14%
H	Installs and Maintains Communications, Networking and Signal Transmission Systems	11%	15%
I	Installs and Maintains Controls Systems	41%	52%
	Total	**100%**	**100%**
In-school theory / practical subject competency weighting		70%	30%

Final in-school percentage score Apprentices must achieve a minimum 70% as the final in-school percentage score to be eligible to write the Interprovincial Red Seal or ITA CofQ exam.	IN-SCHOOL %

All apprentices who complete Level 4 of the Instrumentation and Control Technician (Industrial Instrument Mechanic) program with a FINAL level percentage score of 70% or greater will write the Interprovincial Red Seal examination as their final assessment.

ITA will enter the apprentices' Instrumentation and Control Technician Interprovincial Red Seal examination percentage score in ITA Direct Access.

A minimum percentage score of 70% on the examination is required for a pass.

Appendix B

Glossary and Acronyms

GLOSSARY

Actuator – a controlled hardware device used to implement change in a process

Adapter – a device used to make electrical or mechanical connections between items not originally intended for use together

Align – to bring within required specifications

Amplifier – a device that enables an input signal to control power from a source independent of the signal and thus be capable of delivering an output that bears some relationship to, and is generally greater than, the input signal

Analog signal – any variable signal continuous in both time and amplitude rather than of a pulsed or discrete nature

Apply – to put to use especially for some practical purpose

Back-up – to save configuration, current data or status in recoverable media

Bellows – a mechanical element of generally cylindrical shape with cylindrical walls containing deep convolutions

Benchtest – removing a piece of equipment and testing it at the shop; a static setup as opposed to a dynamic setup

Calibrate – to determine, by measurement or comparison with a standard, the correct value of each scale reading on a meter or other device

Cascade control – a type of controller set-up in which the output of one controller acts as the set point or controlling signal of another controller

Configure – to set up a program or computer system for a particular application

Control mode – a specific type of control action such as proportional, integral or derivative

Control variable – measured variables that can be manipulated by the control system, such as flow, level, pressure and temperature

Describe – to give a detailed or graphic account of a process or procedure

Determine – to arrive at, or locate, information by a process

Distributed Control System (DCS) – a system of dividing plant or process control into several areas of responsibility, each managed by its own controller (processor), with the whole interconnected to form a single entity usually by communication buses of various kinds

Document – to provide proof or evidence

Examine – to investigate critically; scrutinize; test; question

Feed forward – an industry standard process control strategy, in which mathematically predicted errors are corrected before they occur

Fieldbus – a digital, two-way, multi-drop communication link among intelligent measurement and control devices which serves as a Local Area Network (LAN) for advanced process control, remote input/output and high speed factory automation applications; a communication protocol

Firmware – software (programs or data) that has been written onto read-only memory chips; firmware is a combination of software and hardware

Flume – a device that measures large flow rates in open channels

Frequency – the number of cycles completed by a periodic quantity on a unit time

Highway Addressable Remote Terminal (HART) – provides digital communication to microprocessor-based (smart) analog process control instruments; a communication protocol

Human Machine Interface (HMI) – the graphical display and control interface between a process & a human operator

Implement – to make active or effective

Input/Output (I/O) – all equipment and activity that transfers information into or out of a computer

Install – to set up for use or service

Instrumentation – a collection of instruments or their application for the purpose of observation, measurement or control

Instrumentation, Systems and Automation Society (ISA) – an engineering society that develops and maintains defined standards for both scientific and technical areas of process control and automation

Interface – the place at which systems, such as a computer and a peripheral, meet and interact with each other

Kinetic – the energy that a body possesses as a result of its motion

Maintain – to keep in good condition; to keep functional and in good repair

Management of change (facility change management) – proper management of change to industrial facilities and processes is recognized as critical to even small changes; the main requirement is that a thorough review of a proposed change be performed by a multidisciplinary team to ensure that as many possible viewpoints as possible are used to minimize the chances of missing a hazard

Module – an assembly of interconnected components which constitutes an identifiable device, instrument or piece of equipment — can be removed, tested as a unit and replaced with a spare

Network – the interconnection of devices sharing a communications protocol

Operate – to perform a function; exert power or influence

Port – a signal input (access) or output (egress) point

Power supply – a device that produces one or more voltages for the operation of electronic and logic devices

Process – physical or chemical change of matter or conversion of energy such as change in pressure, temperature, speed, electrical potential, etc.

Profibus – a communication protocol

Program – a list of instructions that a computer will execute to perform a certain task

Programmable Logic Controller (PLC) a control device, normally used in industrial control applications, that employs the hardware architecture of a computer and a relay ladder diagram language

Proportional, Integral, Derivative (PID) proportional gain, integral action time and derivative action time. PID software, for example, compares an analog input value with a set point and if there's a discrepancy outputs an appropriate analog or digital control value, according the PID calculations

Range – the region between the limits within which a quantity is measured, received or transmitted; expressed by stating the lower and upper range values

Remote – a device allowing the set point to be altered by a signal from a physical location away from the controller — necessary for cascade operation

Safety Integrity Level (SIL) – Safety Instrument System (SIS)/Process Safety System (PSS)

Sensing element – the element directly responsive to the value of the measured variable

Service – to remove, maintain, repair, or replace items and/or components

Signal – a form of energy that quantitatively represents a variable

Strain gauge – a device that uses the change of electrical resistance of a wire under strain to measure applied force

Supervisory Control and Data Acquisition (SCADA) – a control package used to monitor and control a remote process; also includes hardware such as modems, telemetry, servers and control systems

Telemetry – transmitting the readings of instruments to a remote location via wires, radio waves or other means

Temperature bath – a volume of a substance held at constant temperature, so that an object placed in thermal contact with it is maintained at the same temperature

Terminal – a peripheral device used by the operator to communicate with the computer

Test – to methodically assess against criterion or standard

Thermocouple – devices that convert heat energy into electrical energy consisting of two dissimilar metal strips fused together at one end

Transducer – an element or device that receives energy in one form and converts to another form

Transmitter – a transducer which responds to a measured variable by means of a sensing element, and converts it to a standardized transmission signal that is proportional to the measured variable

Troubleshoot – to investigate critically and methodically the causes of abnormal conditions

Tuning – adjustment of parameters to optimize a particular process

Uninterruptible Power Supply (UPS) used to keep critical equipment, including computers, running in the event of a power failure

Update – to record current data or status

Use – the act or practice of employing something

Variable Frequency Drive (VFD) and Variable Speed Drive (VSD) electronic equipment that allows an electric motor to be run at varying speeds

Weir – an engineered obstruction placed in an open channel

ACRONYMS

ASME – American Society of Mechanical Engineers

A/D, ADC – Analog to Digital Converter

BMS – Burner Management Systems

CEC – Canadian Electrical Code

CEMS – Continuous Emissions Monitoring System

CNSC – Canadian Nuclear Safety Commission

CSA – Canadian Standards Association

CRT – Cathode Ray Tube

D/A, DAC – Digital to Analog Converter

DCS – Distributed Control System

DP – Differential Pressure

EPA – Environment Protection Act

ESD – Emergency Shutdown Device

HART – Highway Addressable Remote Transducer

HMI – Human Machine Interface

I/O – Input/output

ISA – Instrumentation, Systems and Automation Society

LNG – Liquefied Natural Gas

LCD – Liquid Crystal Display

LED – Light Emitting Diode

MISA – Municipal Industry Strategy for Abatement

MSDS – Material Safety Data Sheets

NIST – National Institute of Standards and Technology

OH&S – Occupational Health and Safety Act

OPC – OLE (Object Linking Embedding) Process Control

ORP – Oxidation Reduction Potential

P&ID – Piping & Instrument Drawing

PID – Proportional, Integral, Derivative

PLC – Programmable Logic Controller

PPE – Personal Protection Equipment

PSS – Process Safety Systems

RTU – Remote Terminal Unit

RTD – Resistive Temperature Device

SAC – Stand Alone Controller

SCADA – Supervisory Control and Data Acquisition

SIL – Safety Integrity Level

SIS – Safety Instrument System

SOP – Standard Operating Procedures

TCP/IP – Transport Control Protocol / Internet Protocol

TDG – Transportation of Dangerous Goods

UPS – Uninterruptible Power Supply

VFD – Variable Frequency Drive

VSD – Variable Speed Drive

WHMIS – Workplace Hazardous Materials Information System

Appendix C

Sample Evaluation Sheet

SAMPLE EVALUATION SHEET

INSTRUMENTATION AND CONTROL TECHNICIAN (INDUSTRIAL INSTRUMENT MECHANIC)

Date: _____

Practical Skills Assessment Form

Student _____ **Score:** _____ **%**

Lab Project:_____

School/Training Institution: _____

Instructor/ Assessor: _____

SAMPLE

General Work Standard Criteria **Rate 1-10 or N/A**

1. Plans Lab tasks to organize work and ensure personal safety ……. _____
2. Uses required tools safely and efficiently…… ……_____
3. Maintains orderly work area while completing Lab assignment…… _____
4. Leaves Lab work area in a clean, work-ready state …… _____

Lab Criteria **Rate 1-10 or N/A**

1. Demonstrates knowledge of related theory and principles ……. _____
2. Interprets related schematics or drawings correctly…….. _____
3. Demonstrates understanding of primary device or system…….. _____
4. Demonstrates understanding of secondary device or system……. _____
5. Efficiently manipulates devices or systems…….. _____
6. Solution to Lab assignment is valid within stipulated parameters… _____
7. Documents calibration as required ……. _____
8. Completes Lab assignment within time allotted …….. _____
9. Meets other Lab criteria (Identify:_____) ……. _____
10. Meets other Lab criteria (Identify:_____) ……. _____

Scoring passing grade is 70%:

- ***Add Total scores for all criteria,***
- ***divide by total number of applicable criteria,***
- ***multiply by 100%***

NOTE: It is recommended that at least one Practical Assessment be completed for each line on the Occupational Analysis Chart (e.g. General Area of Competency) for each term.

Appendix D

Previous Contributors

Previous Contributors

The Program Outline was prepared under the direction of an Industry Steering Committee convened by the Resource Training Organization (RTO). Members include:

- Curt Cain — RTO
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